

LECTURE MODULE · QUANT FINANCE ON THE IDX

# Reading the IDX Quant Monitor

How to read every chart on the dashboard, tab by tab: what it shows, how to read it, and when it should change a decision.

9 tabs · 40+ charts and tables

Data to 6 Oct 2026

For learning, not investment advice

## THE METHOD

# Four questions to ask of any chart

1

### What is on each axis?

Rupiah, percent,  $\sigma$ , a rank from 0 to 100? The unit decides what “big” means.

2

### Compared with what?

IHSG, the zero line, the stock’s own past, or the other 98 stocks. No benchmark, no meaning.

3

### Where is “now”?

The tag on the right edge or the highlighted row. History explains; the last point decides.

4

### What decision changes?

Buy, hold, trim, resize, or nothing. If no decision changes, the chart is background.

THE MAP

# Each tab answers one investor question

| Tab           | Question it answers                         | Main charts                                                           |
|---------------|---------------------------------------------|-----------------------------------------------------------------------|
| Portfolio     | What do I own, and is it beating the index? | Headline numbers, performance, attribution, my indices                |
| Risk & return | How much risk do I carry, and is it paid?   | Sharpe, risk contribution, VaR, risk vs momentum, volatility          |
| Market        | What are the market and the world doing?    | IHSG outlook, macro odds, relative performance, breadth, foreign flow |
| Stocks        | Where does each stock sit among the 99?     | Golden Line explorer, alpha vs beta map, 3D map                       |
| Stock         | What is happening in one company?           | Price, flow, entry signal, Monte Carlo, beta, bell curve, pairs       |
| Ideas         | What does the formula say to buy today?     | Stock picks for V1+, V2, V3, Golden v2+, Golden A+                    |
| Allocation    | How much should go into each stock?         | Allocation plan, efficient frontier                                   |
| Backtest      | Would the rules have worked?                | Formula comparison, Kelly · Bayes · EV, v3, pattern lab, segments     |

The Settings tab holds data upload and account options only, so it has no charts to read.

TAB 01 OF 08

# Portfolio

*What do I own, and is it beating the index?*

Headline numbers & positions

Portfolio performance

P&L attribution

My indices

# Headline numbers and positions

Market value

IDR 471.7M

cost IDR 456M · 5 positions

Unrealised P&L

+IDR 15.7M

+3.43% · net of fees +IDR 13.8M

vs IHSG same period

+2.2 pp

IHSG would be +IDR 5.5M

VaR 1D hist 95

IDR 12.8M

2.71% of NAV

CVaR 1D hist

IDR 19M

4.03% of NAV

Beta IHSG

0.82

1% IHSG ≈ IDR 3.9M

Stress: all ARB

IDR -70.7M

one session

Positions

edit lots, avg price, entry date inline · blank price = close on entry date

TICKER

2000

lots

avg price

mm/dd/yyyy

+ Add

Clear all

| Ticker | Lots | Avg price | Entry date | Last   | Value IDR | P&L IDR | P&L %  | vs IHSG  | Weight | Beta | Risk share | Key point                                                                                                                                                                 |
|--------|------|-----------|------------|--------|-----------|---------|--------|----------|--------|------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AADI   | 100  | 11000     | 08/20/2026 | 12,125 | 121.3M    | +11.3M  | +10.2% | +15.0 pp | 25.7%  | 0.43 | 25.7%      | <div>STRONG 4/4</div> P&L +10.2% (+15.0 pts vs IHSG) · Defensive Compounders · uptrend · foreign 20D +IDR 378 bn · <div>STOP HIT EARLIER</div> · stop 11,650...           |
| ANTM   | 300  | 2700      | 08/03/2026 | 3,250  | 97.5M     | +16.5M  | +20.4% | +21.0 pp | 20.7%  | 1.18 | 27.6%      | <div>STRONG 4/4</div> P&L +20.4% (+21.0 pts vs IHSG) · Aggressive Leaders (α not significant) · uptrend · foreign 20D +IDR 663 bn <div>▲ risk 28% &gt; weight 21...</div> |
| BBRI   | 500  | 2900      | 07/01/2026 | 3,110  | 155.5M    | +10.5M  | +7.2%  | -1.5 pp  | 33.0%  | 0.82 | 25.3%      | <div>MIXED 2/4</div> P&L +7.2% (-1.5 pts vs IHSG) · Defensive Compounders (α not significant) · mixed trend · foreign 20D -IDR 1,042 bn · <div>EXIT</div> · stop 3,220... |
| GOTO   | 2000 | 60        | 05/04/2026 | 31     | 6.2M      | -5.8M   | -48.3% | -37.2 pp | 1.3%   | 0.47 | 0.3%       | <div>WEAK 0/4</div> P&L -48.3% (-37.2 pts vs IHSG) · Defensive Laggards · downtrend · foreign 20D -IDR 1,351 bn <div>▲ RSI 18 oversold</div> · drawdown...                |
| TLKM   | 400  | 2700      | 06/02/2026 | 2,280  | 91.2M     | -16.8M  | -15.6% | -15.5 pp | 19.3%  | 0.99 | 21.0%      | <div>WEAK 0/4</div> P&L -15.6% (-15.5 pts vs IHSG) · Defensive Laggards (α not significant) · downtrend · foreign 20D -IDR 1,094 bn <div>▲ RSI 28 oversold</div> · ...    |
| Total  |      |           |            |        | 471.7M    | +15.7M  | +3.4%  | +2.2 pp  | 100.0% | 0.82 | 100.0%     |                                                                                                                                                                           |

## SHOWS

Seven headline numbers, then one row per holding: P&L, gap vs IHSG, weight, beta, risk share.

## READ IT

Start with “vs IHSG” (+2.2 pp), then compare each row’s Weight with its Risk share.

## ACT WHEN

A WEAK 0/4 label, an EXIT or a stop hit, or risk share well above weight: review that row first.

**Trap:** “Stress: all ARB” is a one-day worst case if every stock hits its lower limit, not a forecast.

# Portfolio performance vs the same money in IHSG



### SHOWS

Brown = your unrealised P&L in rupiah.  
Dotted grey = the same money in IHSG on the same dates.

### READ IT

The gap between the two lines is your edge. Green fill = in profit, red fill = under water.

### ACT WHEN

Brown stays under grey for weeks: your picks are costing money versus simply owning the index.

**Trap:** a rising line in a rising market is not skill. Always read it against the dotted line.

# P&L attribution: which positions made the money



**SHOWS**

One bar per holding, in rupiah, sorted from the biggest gain to the biggest loss.

**READ IT**

ANTM, AADI and BBRI made money; TLKM lost the most. “3/5 in profit” sits in the corner.

**ACT WHEN**

One bar dominates the total: the result rests on one stock. Was that the plan?

**Trap:** rupiah mixes size and skill. A big bar can be a big position with a small move.

# My indices: your own benchmark at 100



SHOWS

IHSG, today’s portfolio and any basket you build, all starting at 100 one year ago.

READ IT

Above 100 = gained since the start. The table adds volatility, max drawdown, Sharpe, beta.

ACT WHEN

Build a peer basket (e.g. big banks) to test a stock against its own sector, not only IHSG.

Trap: “My portfolio” holds today’s lots all year (buy & hold), not your real trading history.

# Risk & return

*How much risk do I carry, and am I paid for it?*

Sharpe ratio

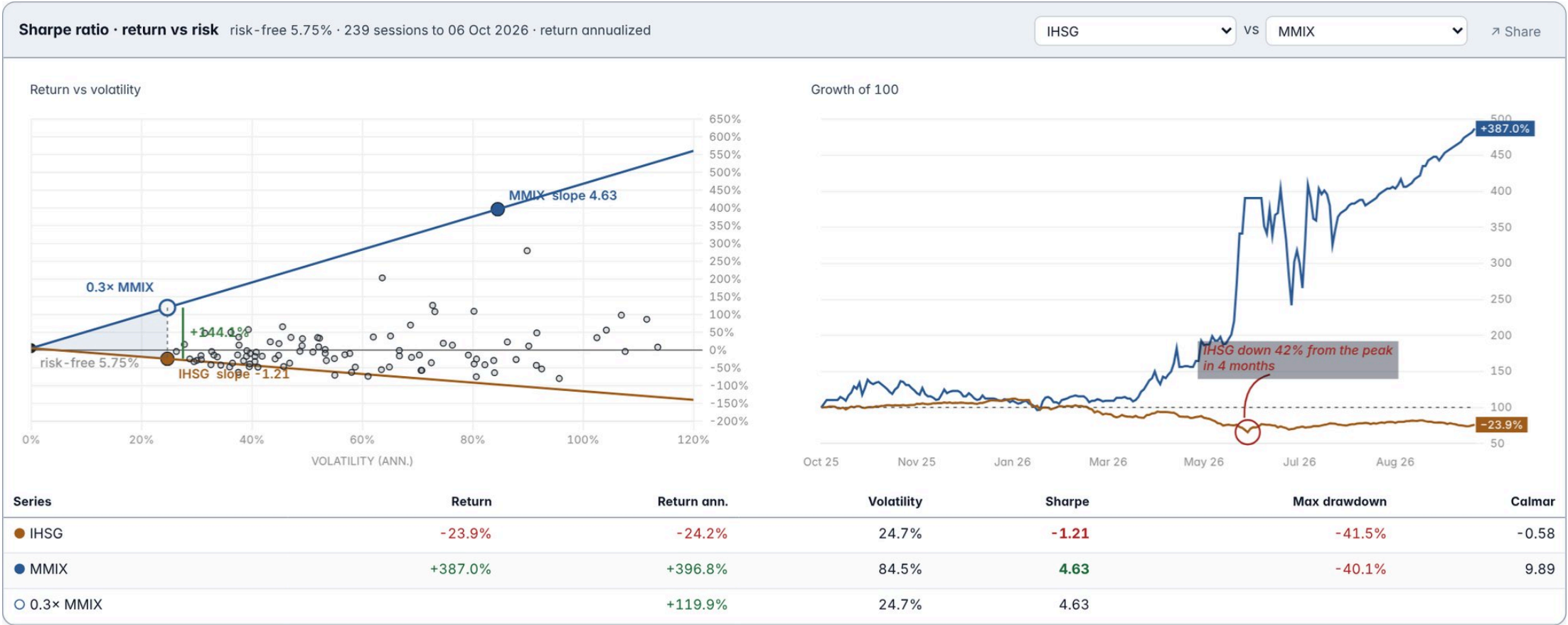
Risk contribution

P&L vs VaR

Risk vs momentum

Volatility, a day to a year

# Sharpe ratio: return per unit of risk



SHOWS

Left: stocks by volatility (across) and return (up). Right: growth of 100 for the chosen pair.

READ IT

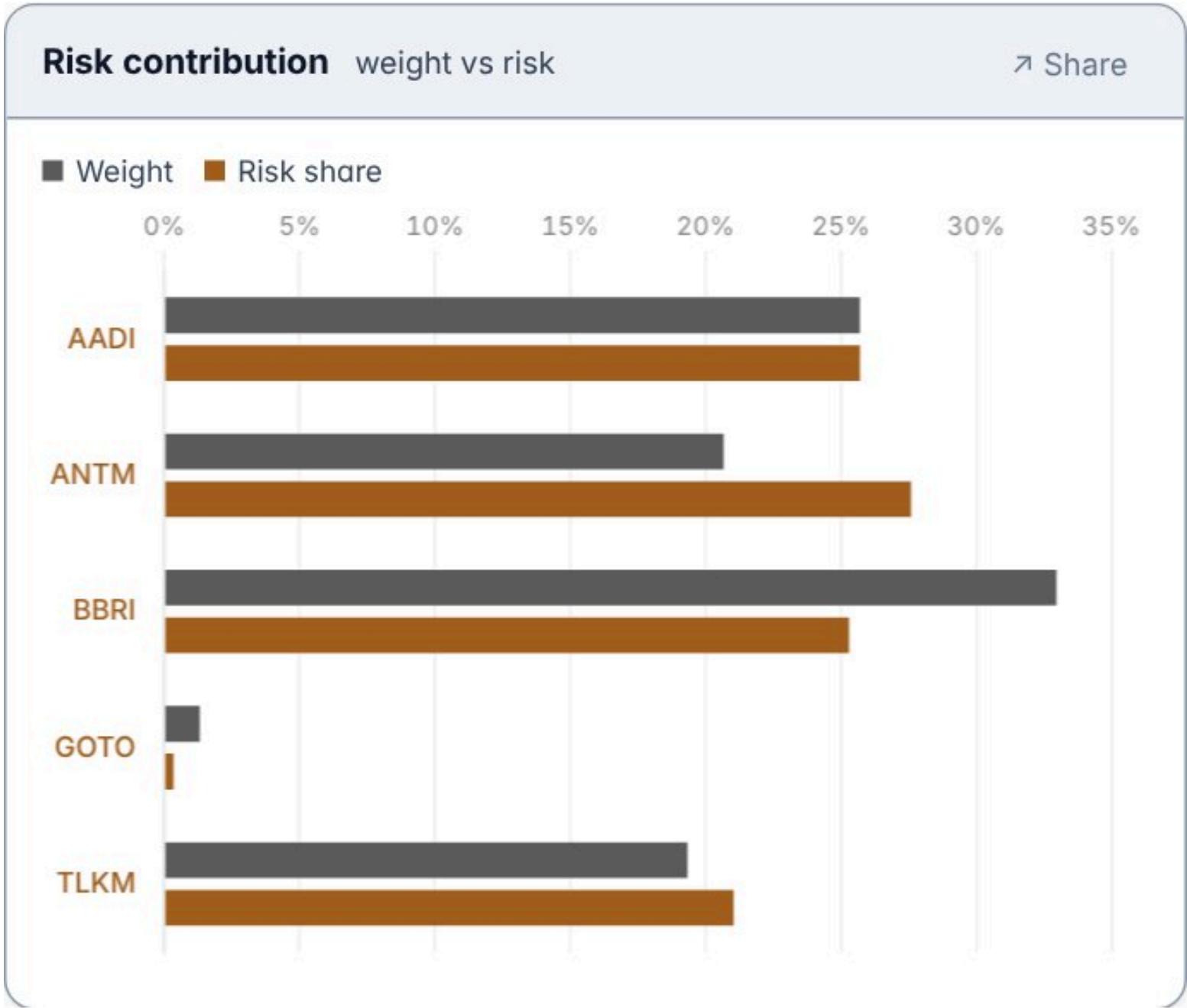
The line from risk-free (5.75%) to a stock: its slope is the Sharpe ratio. Steeper is better.

ACT WHEN

Compare candidates by slope, not by return. “0.3× MMIX” = MMIX scaled to IHSG’s risk.

**Trap:** one great year is not a Sharpe you can count on. MMIX also fell 42% in four months.

# Risk contribution: money share vs risk share



**SHOWS**

Grey = share of your money in each stock. Brown = share of total portfolio risk it causes.

**READ IT**

Brown longer than grey = more risk than its size.  
ANTM: 21% of the money, 28% of the risk.

**ACT WHEN**

Trim stocks whose brown bar is far longer than grey, or add calmer names to balance.

**Trap:** risk share depends on correlation. Two stocks that move together count almost double.

# P&L vs VaR: how bad is a bad day?



**SHOWS**

Daily % change of today’s portfolio over 239 days. Dashed line = VaR 95% at -2.71%.

**READ IT**

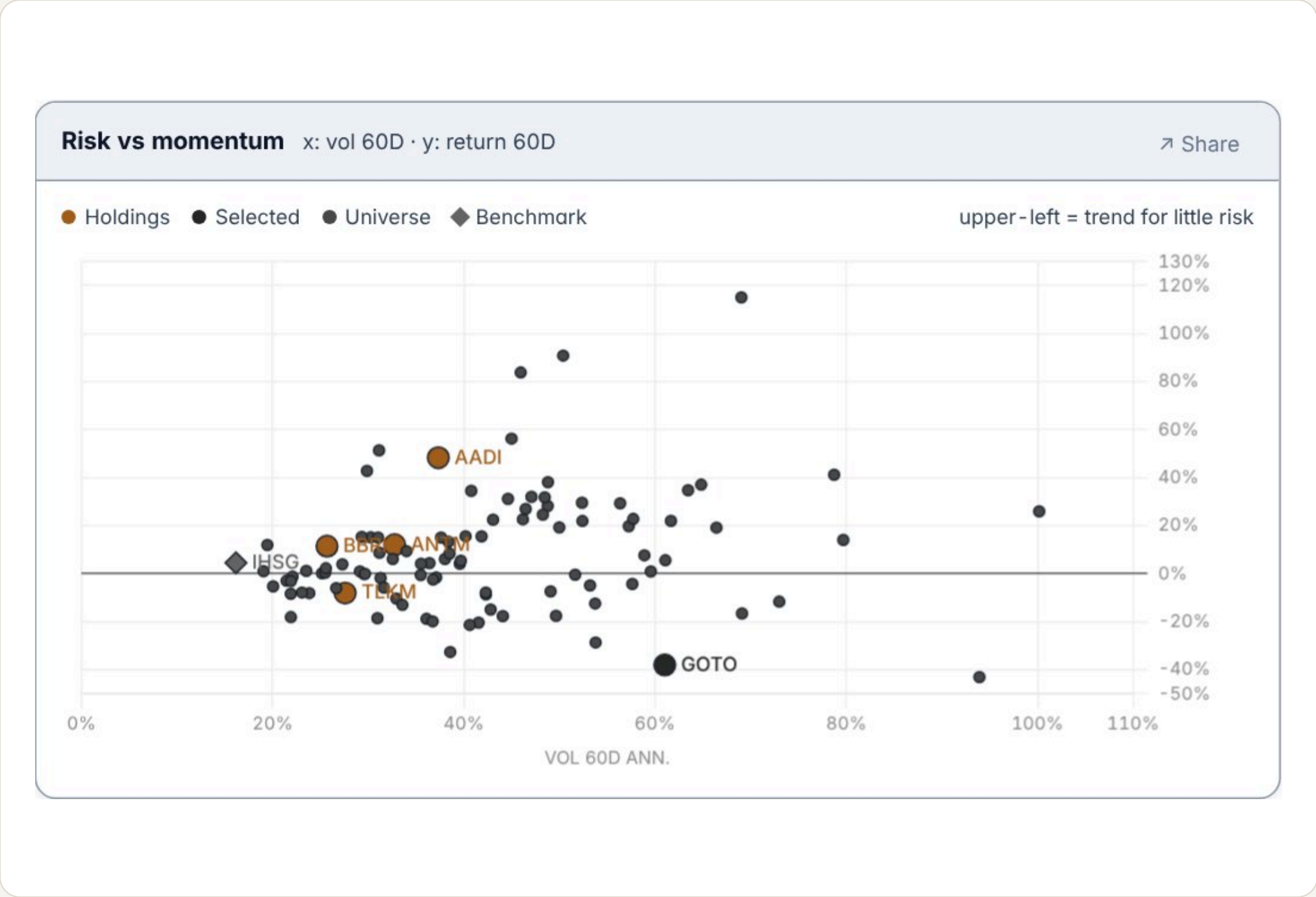
About 1 day in 20 should be worse than the line. Dark red bars are those breaches: 12.

**ACT WHEN**

Expected ≈ 12 (5% of 239). Many more breaches = the risk model is too optimistic: cut size.

**Trap:** VaR says how often, not how much. On a breach day the loss can be -7%, far past the line.

# Risk vs momentum: is the trend worth it?



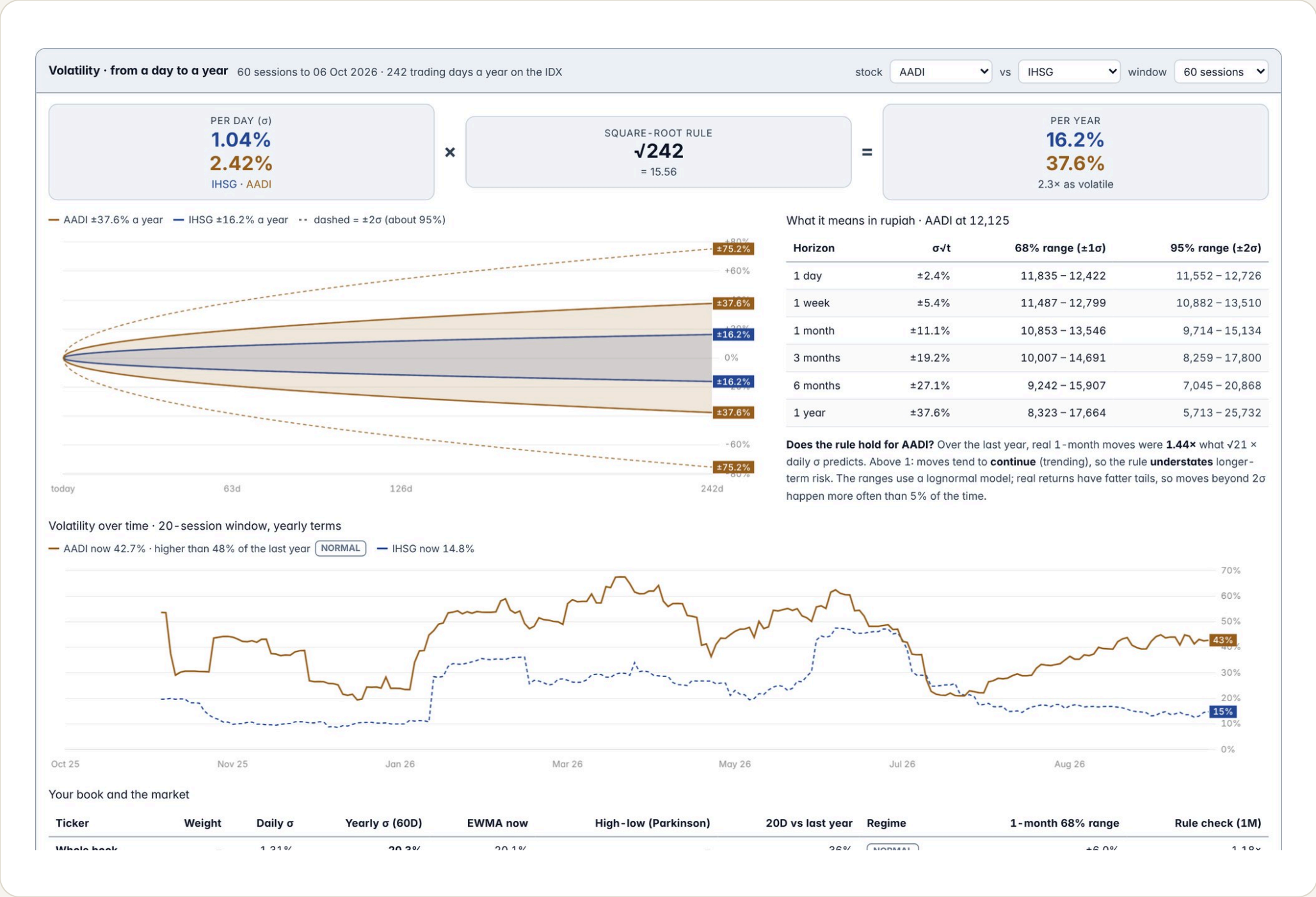
**SHOWS**  
Each dot is a stock: 60-day volatility across, 60-day return up. Brown = holdings, ◆ = IHSG.

**READ IT**  
Upper-left is best: a strong trend with little risk. Lower-right is worst: falling and wild.

**ACT WHEN**  
A holding in the lower-right corner deserves a review, a tighter stop or a smaller size.

**Trap:** 60 days is short. One news event can move a stock to another corner.

# Volatility: from one day to one year



# Market

*What are the market and the world around it doing?*

IHSG probability outlook

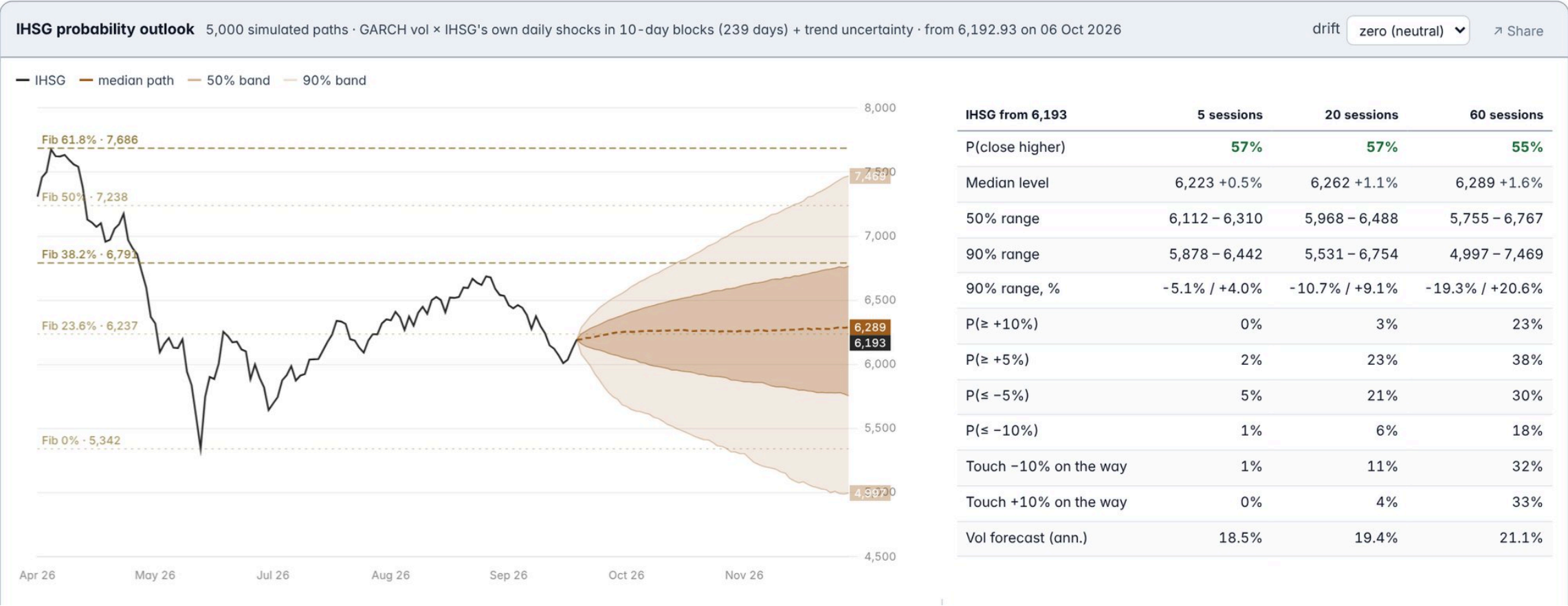
Macro odds (Kalshi)

Relative performance

Market breadth

Foreign flow leaders

# IHSG outlook: a range, not a target



SHOWS

5,000 simulated paths built from IHSG’s own past shocks. Bands = the 50% and 90% ranges.

READ IT

From 6,193: 57% chance higher in 20 sessions; 90% range 5,531–6,754 (–10.7% / +9.1%).

ACT WHEN

Stress-test your book with the 90% band. Check any price target against the target table.

Trap: 57% is barely better than a coin flip. The value is in the width of the range, not its direction.

# Macro odds: what real-money markets expect

SHOWS

Kalshi prices: 84¢ for a \$1 contract = 84% odds.  
Fed, recession, CPI and the S&P 500.

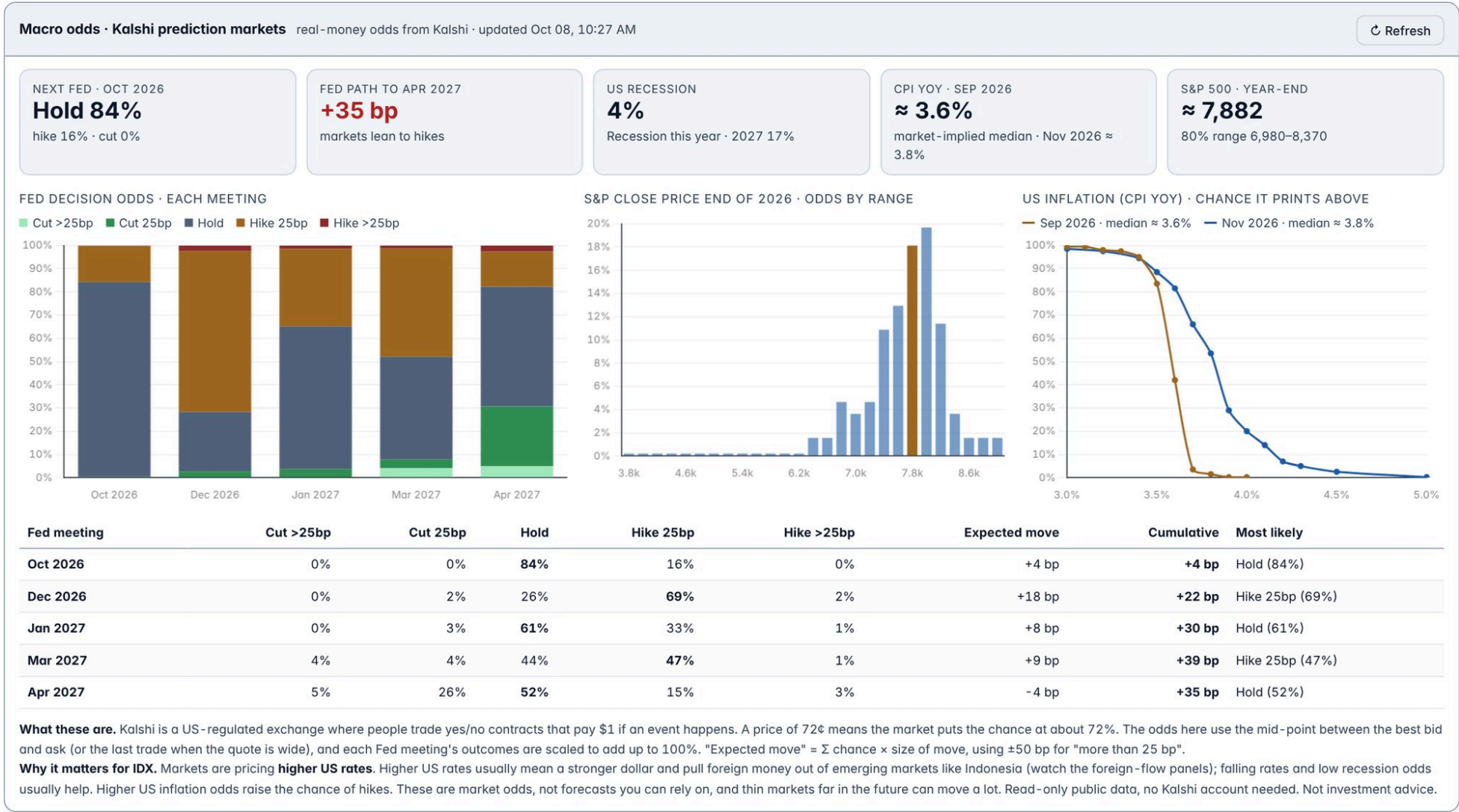
READ IT

Next Fed: hold 84%, hike 16%. Path to April 2027 +35 bp: markets lean to higher US rates.

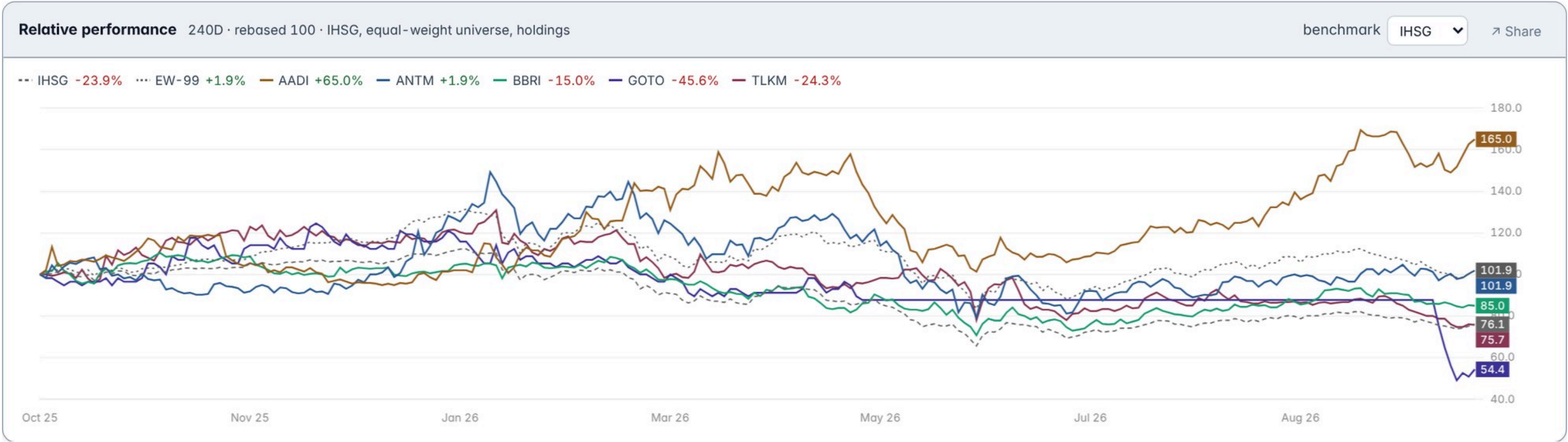
ACT WHEN

Rising hike odds = stronger dollar, foreign selling of EM stocks: check foreign flow.

**Trap:** these are US events. They reach the IDX only via rates, the rupiah and flows.



# Relative performance: who led the year



## SHOWS

Every line starts at 100: IHSG, the equal-weight 99 stocks (EW-99) and your holdings.

## READ IT

IHSG -23.9% but EW-99 +1.9%: the average stock beat the big caps that drive the index.

## ACT WHEN

When EW beats IHSG, picking stocks beyond the largest caps has room to add value.

**Trap:** rebasing hides the path. Two lines can end at the same level after very different drawdowns.

# Market breadth: how many stocks are rising



**SHOWS**

Share of the 99 stocks above their 50-day (brown) and 20-day (blue) average price.

**READ IT**

Above 80% = almost all rising (stretched). Below 20% = almost all falling (washed out).

**ACT WHEN**

Breadth turning up from below 20% (now 28% / 22%) is an early sign of broad recovery.

**Trap:** IHSG up while breadth falls = a narrow rally on a few big caps. Fragile.

# Foreign flow leaders: where foreign money goes



**SHOWS**  
Net foreign buying (green) and selling (red) per stock over 5, 20 or 60 days, IDR billion.

**READ IT**  
20 days: foreigners net sold IDR 12.5 tn of IHSG. They bought ANTM, AADI; sold TLKM, BBRI.

**ACT WHEN**  
Heavy, repeated selling in a stock you hold is supply pressure. Read it with the price trend.

**Trap:** one large block trade can dominate a bar. Check 5D and 60D too.

TAB 04 OF 08

# Stocks

*Where does each stock sit among the 99?*

Golden Line explorer

Alpha vs Beta map

3D stock map

# Golden Line explorer: today’s chart signals

Golden Line explorer

your AFL Explorer: value ≥ 5 bn today and 20-day average ≥ 10 bn · 81 stocks · 06 Oct 2026 · click a row to open

min value5bnAll calls

| Ticker | Close  | Pattern                  | Calls    | Allocation | Remark                               | Net foreign (bn) | Resistance | Support |
|--------|--------|--------------------------|----------|------------|--------------------------------------|------------------|------------|---------|
| PTBA   | 3,400  | <div>■ Blossom</div>     | Buy      | above 70%  | Trend strengthening                  | 26.9             | 3,620      | 3,260   |
| AADI   | 12,125 | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | -31.7            | 12,925     | 11,625  |
| ADRO   | 2,600  | <div>■ Blossom</div>     | Buy      | max 50%    | Medium-term trend strengthening      | -9.9             | 2,760      | 2,500   |
| SGER   | 775    | <div>■ Blossom</div>     | Buy      | max 50%    | –                                    | 0.9              | 835        | 740     |
| TOWR   | 428    | <div>■ Blossom</div>     | Buy      | above 70%  | Trend strengthening on rising volume | -18.7            | 456        | 410     |
| KLBF   | 755    | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | 1.4              | 790        | 730     |
| SSIA   | 1,760  | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | 0.9              | 1,870      | 1,690   |
| SMIL   | 472    | <div>■ Blossom</div>     | Buy      | above 70%  | Trend strengthening on rising volume | 2.6              | 505        | 452     |
| UVCR   | 284    | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | 6.2              | 326        | 258     |
| AYAM   | 530    | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | 1.8              | 565        | 510     |
| SCMA   | 181    | <div>■ Blossom</div>     | Buy      | max 50%    | Trend strengthening                  | 1.0              | 192        | 174     |
| GOTO   | 31     | <div>■ Early bloom</div> | Spec Buy | max 20%    | Average up on a strong green candle  | -255.8           | 36         | 50      |
| AMMN   | 4,500  | <div>■ Early bloom</div> | Spec Buy | max 20%    | Average up on a strong green candle  | 19.4             | 4,880      | 4,260   |
| ANTM   | 3,250  | <div>■ Early bloom</div> | Spec Buy | max 20%    | Average up on a strong green candle  | 35.8             | 3,420      | 3,140   |
| ENRG   | 1,080  | <div>■ Early bloom</div> | Spec Buy | max 20%    | Average up on a strong green candle  | 20.9             | 1,185      | 1,015   |
| TINS   | 1,670  | <div>■ Early bloom</div> | Spec Buy | max 20%    | Average up on a strong green candle  | 38.2             | 1,910      | 1,500   |

SHOWS

Each liquid stock’s Golden Line pattern, call, size cap, foreign flow, resistance, support.

READ IT

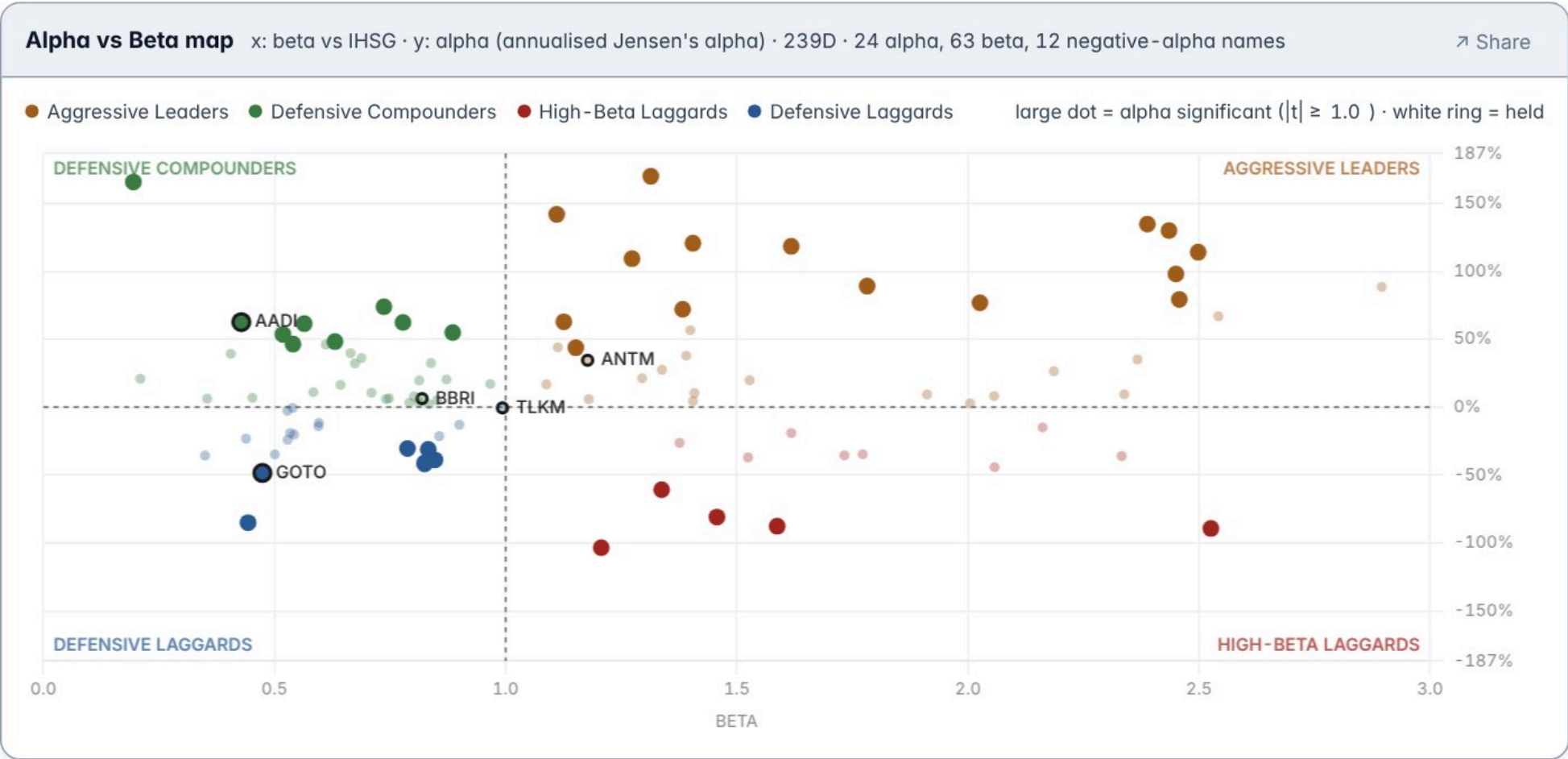
Blossom = confirmed uptrend (Buy). Early bloom = a fresh turn (Spec Buy, max 20% size).

ACT WHEN

Prefer Buy rows with positive foreign flow and a price not far above support.

Trap: the raw chart signal had a weak edge in the backtest. Use it with the anti-chase filter.

# Alpha vs Beta map: four kinds of stock



**SHOWS**

Across: beta (market sensitivity). Up: alpha, the yearly return the market can't explain.

**READ IT**

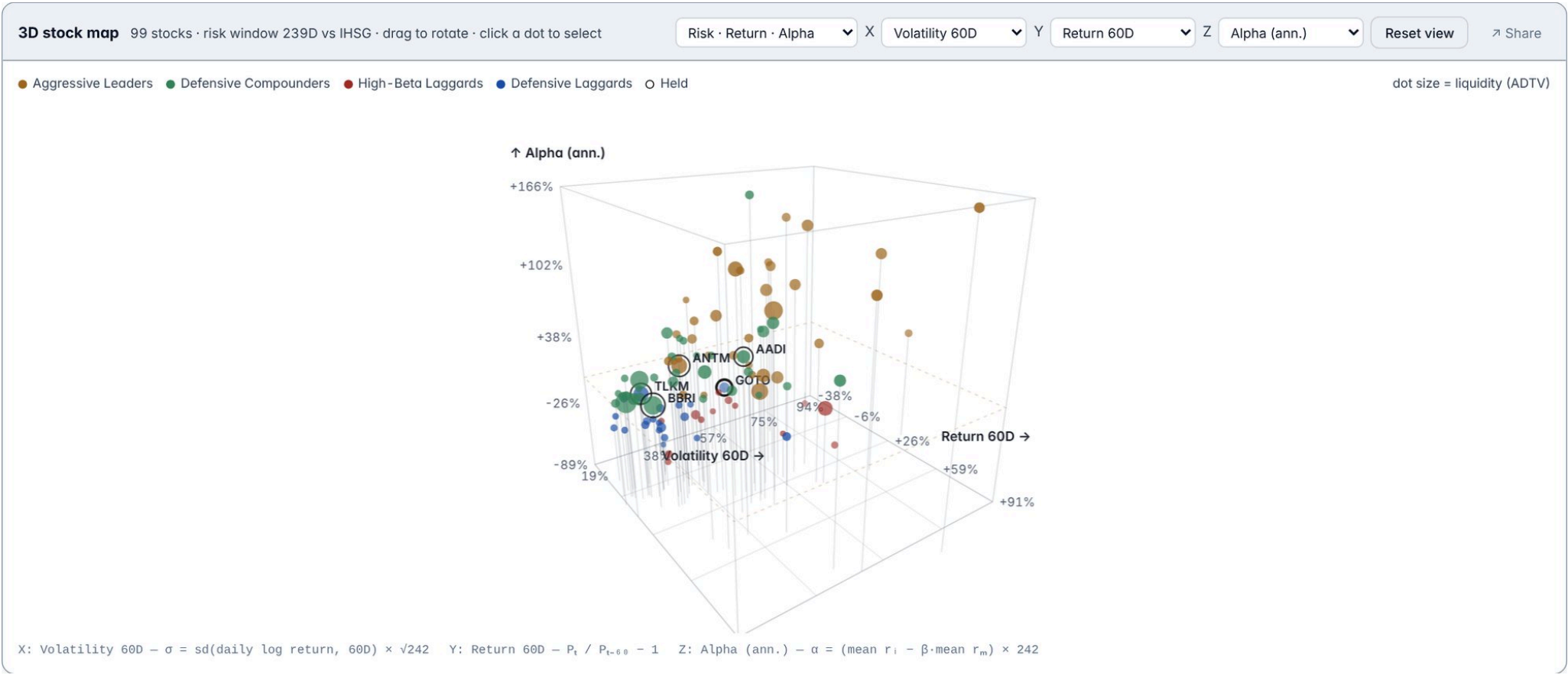
Top-left: Defensive Compounds (low beta, + alpha). Top-right: Aggressive Leaders.

**ACT WHEN**

In weak markets favour the top-left; avoid the bottom-right High-Beta Laggards.

**Trap:** one year of alpha is noisy. Only large dots are statistically clear.

# 3D stock map: three measures at once



## SHOWS

Volatility, 60-day return and alpha on three axes. Colour = segment, dot size = liquidity.

## READ IT

Drag to rotate. A dot that stands apart from the cloud on every axis deserves a look.

## ACT WHEN

Use it to explore, not decide: click a dot to open that stock on the Stock tab.

**Trap:** 3D perspective distorts distance. Confirm what you see on the flat 2D maps.

# Stock

*What is happening in one company? (example: BBRI)*

Price

Foreign flow

Entry signal

Monte Carlo

Volatility & drawdown

Beta

Bell curve

Model view

Pair spread

# Price chart: trend, bands and the stop



**SHOWS**

Candles with 20- and 50-day averages, Bollinger bands, volume, entry and stop lines.

**READ IT**

Header: RSI 41, 2.2% under MA50, EXIT: price is 3.5% below the 3,220 trailing stop.

**ACT WHEN**

Price below its stop: the exit rule wins over any new buy signal. Respect the stop first.

**Trap:** indicators describe the past. The stop is the only line here that is a decision.

# Foreign flow for one stock



**SHOWS**

Daily net foreign buy (green) or sell (red) bars and the running total line, IDR billion.

**READ IT**

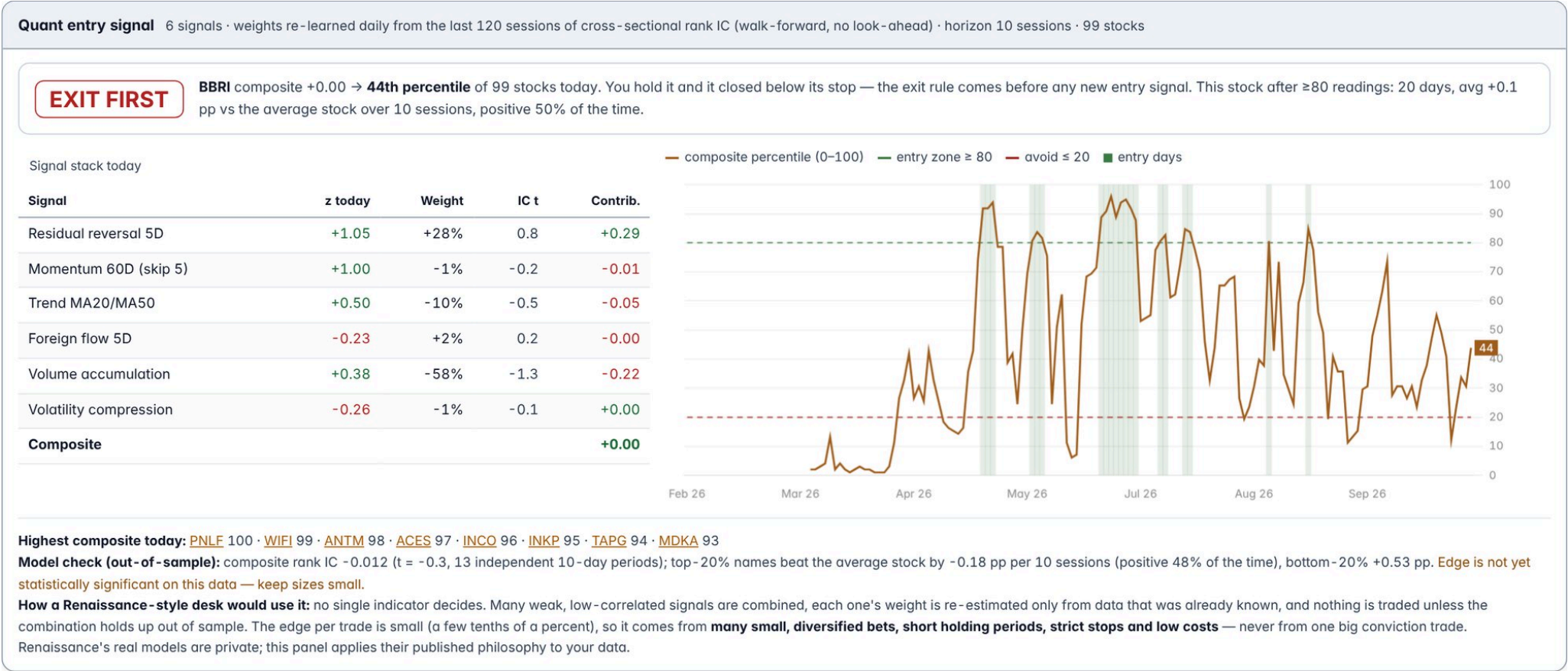
BBRI: −1,042 bn over 20 days, 9.8% of traded value. Foreigners are steady sellers.

**ACT WHEN**

Line and price falling together confirm a downtrend. Line rising while price falls hints at a turn.

**Trap:** the running total has no natural zero. Read its slope, not its level.

# Quant entry signal: six signals, one score



## SHOWS

Six signal scores, each weighted by how well it predicted lately. Their sum is the composite.

## READ IT

Composite percentile over time: above 80 = entry zone, below 20 = avoid. BBRI is at 44.

## ACT WHEN

Enter only in the green zone. EXIT FIRST means a stop was hit: exit before any new buy.

**Trap:** out of sample, the edge is a few tenths of a percent per trade. Size it modestly.

# Monte Carlo: what could happen to this position

## SHOWS

3,000 simulated 20-day paths for IDR 155.5 mn  
in BBRI, built from its own past shocks.

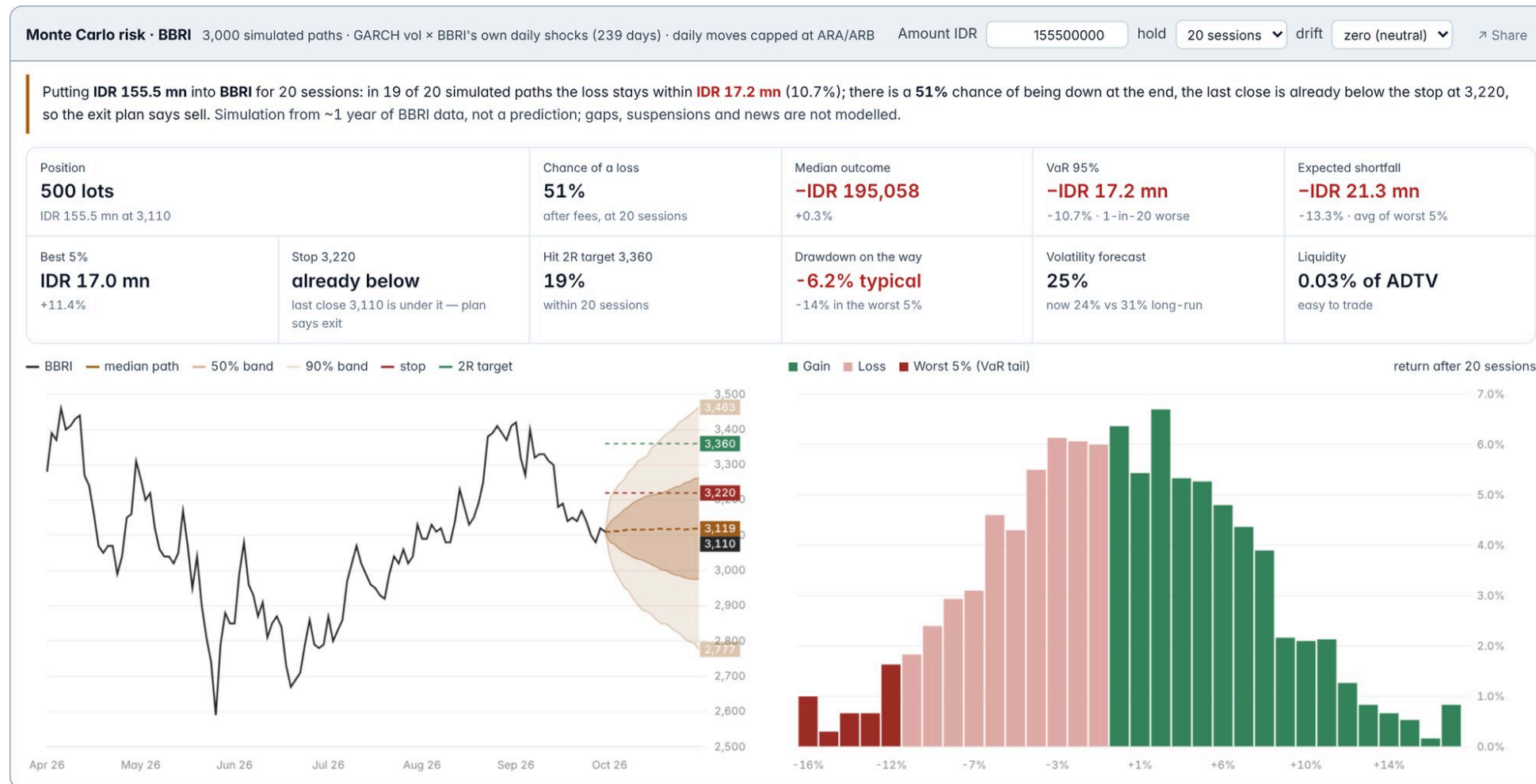
## READ IT

51% chance of a loss. 1 in 20 paths loses over IDR 17.2 mn; their average loss is 21.3 mn.

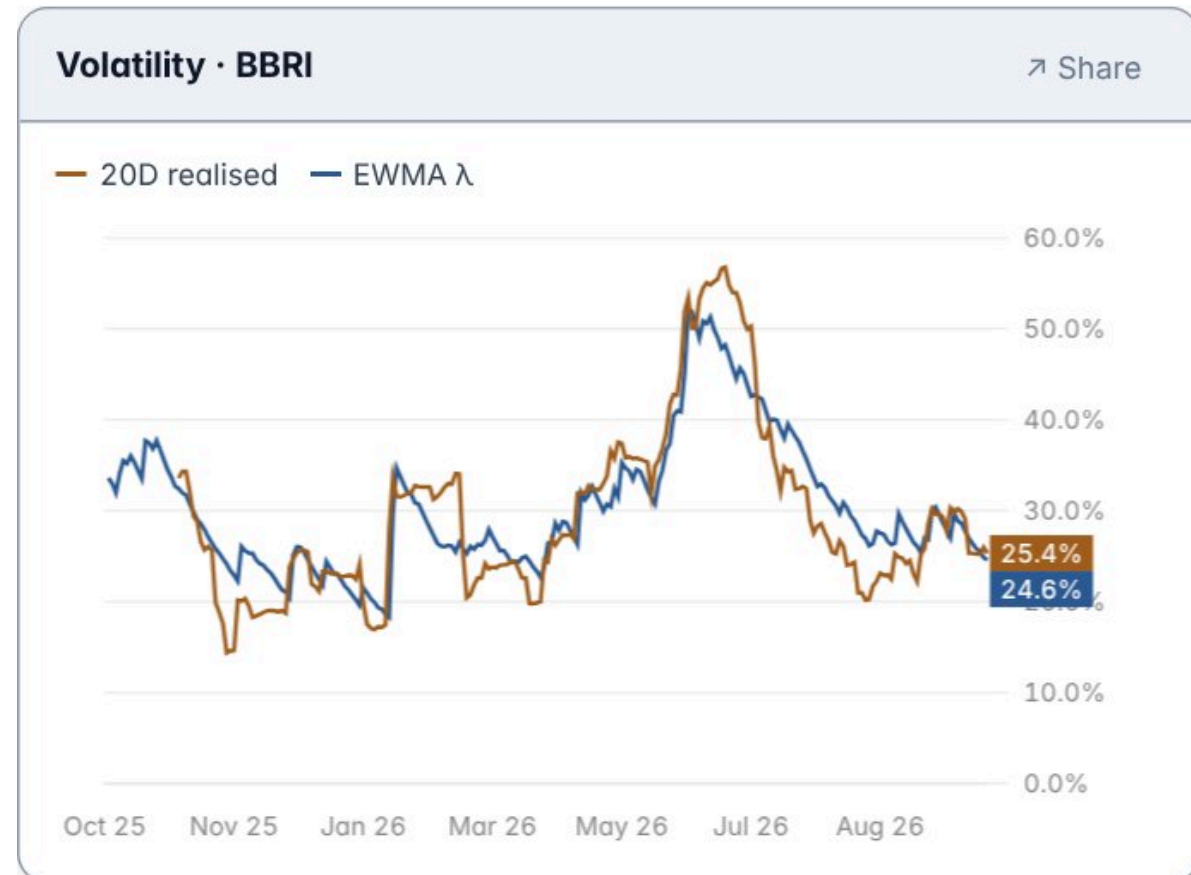
## ACT WHEN

If the 1-in-20 loss is more than you accept, lower the amount and rerun until it fits.

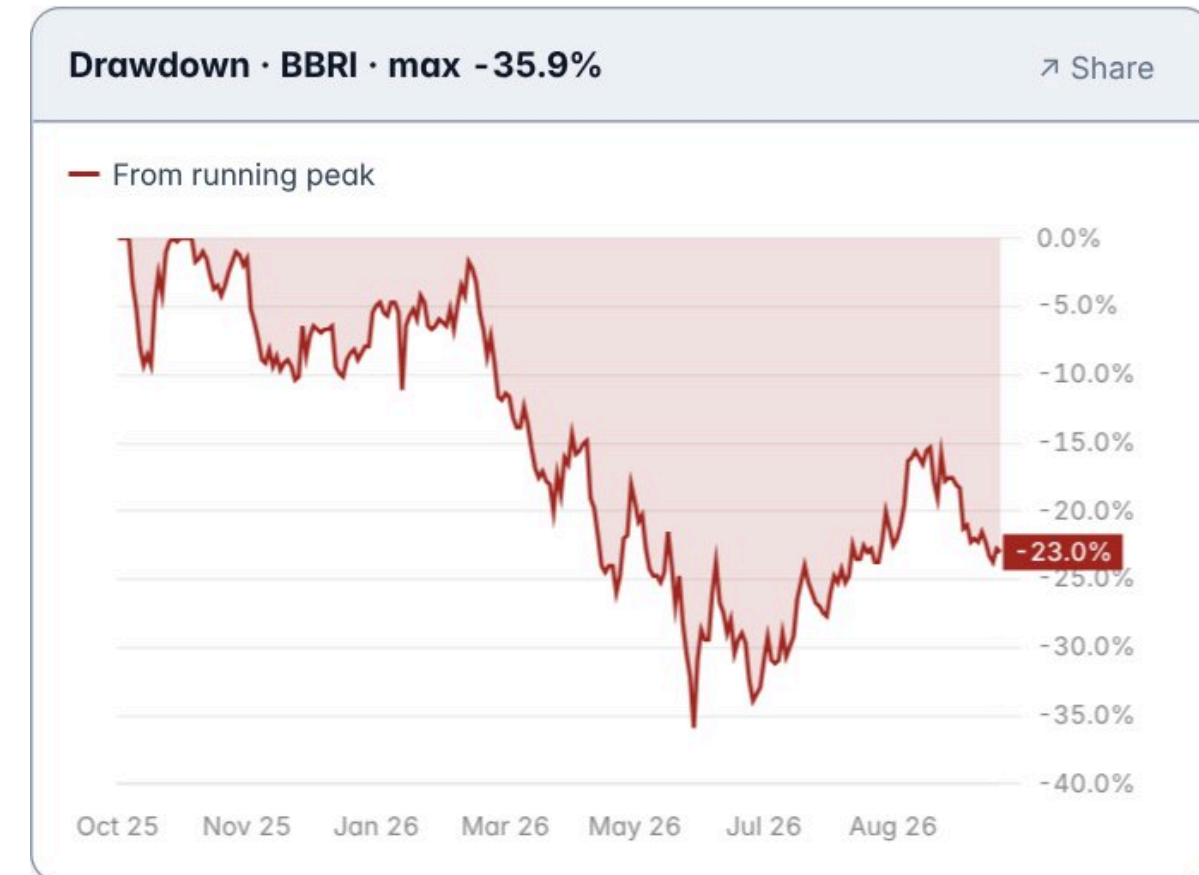
**Trap:** the simulation only knows last year.  
Suspensions and news shocks are not in it.



# Volatility and drawdown: how rough the ride is



**Volatility:** 20-day realised (brown) vs EWMA (blue), per year. EWMA reacts faster to recent days. Rising = wider stops, smaller size.



**Drawdown:** % below the previous peak. BBRI's worst was -35.9%. The deeper the hole, the bigger the gain needed to recover.

**Trap:** a flat 0% volatility line means no trading (e.g. a suspension), not a safe stock.

# Rolling beta and correlation vs IHSG



**SHOWS**

Beta (brown): move per 1% IHSG move.  
Correlation (blue): how closely it follows the index.

**READ IT**

BBRI beta now 1.01 (range 0.36–1.09): it moves about one-for-one with the index.

**ACT WHEN**

Beta rising toward 1 or above in a weak market adds index risk. Size accordingly.

**Trap:** beta with low correlation is unreliable: the stock isn't really following IHSG.

# The bell curve: why most days are ordinary

SHOWS

Left: 12 coin flips; most paths land near 6 up.  
Right: BBRI’s real daily returns vs the bell.

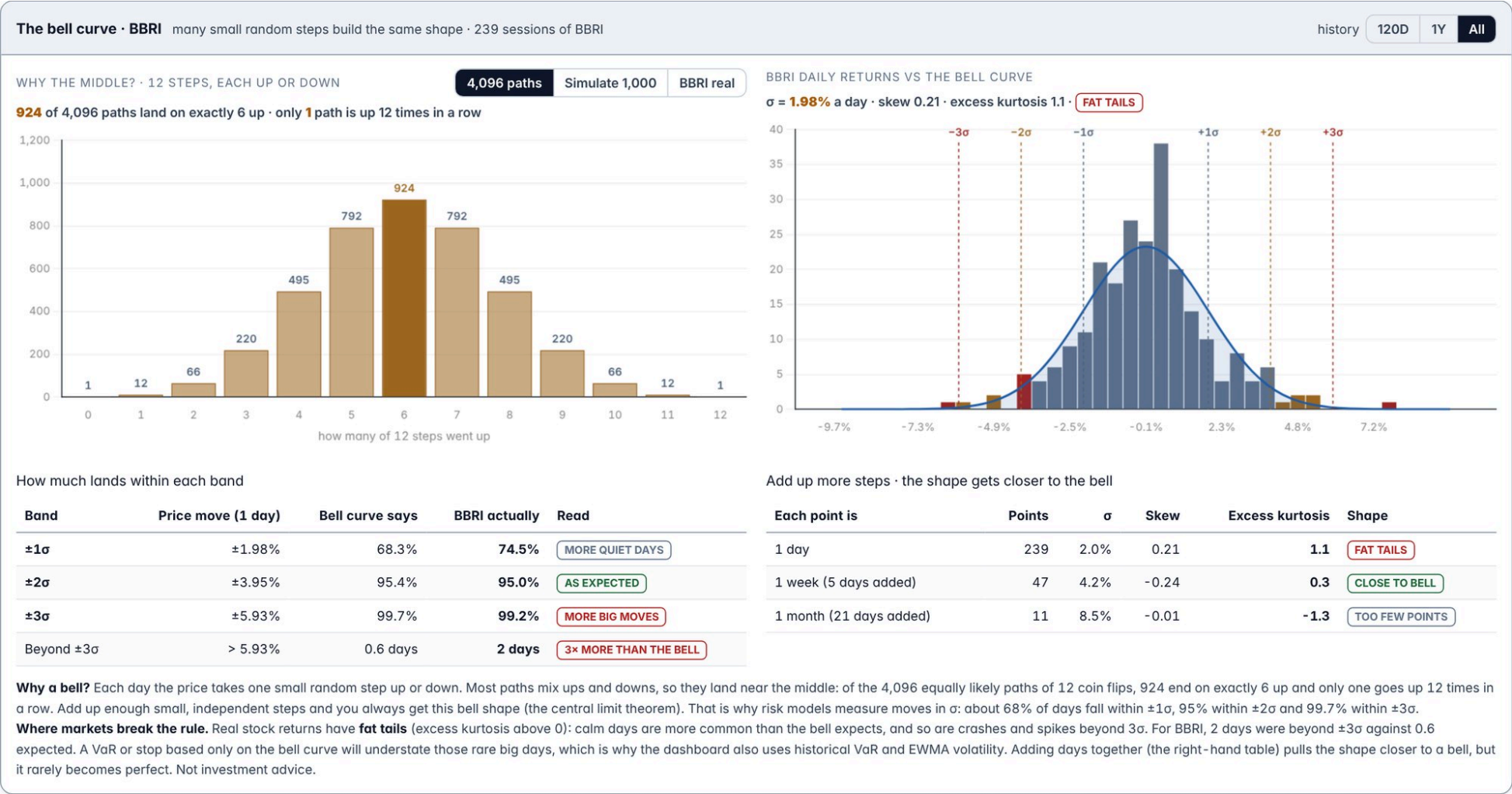
READ IT

$\sigma = 1.98\%$  a day. The table counts real days in each band: more big days than the bell says.

ACT WHEN

Plan for the tails: size so that a  $3\sigma$  day (about  $-6\%$  here) is survivable.

**Trap:** the bell alone underestimates crashes.  
That is why VaR uses real history.



# Model view: why the model rates a stock



**SHOWS**

Grey = each factor’s z-score vs all stocks. Brown = what that factor adds to the final score.

**READ IT**

BBRI: low volatility and 5-day reversal lift the score; foreign flow drags it. Rank 35/97.

**ACT WHEN**

Check the reasons match your view of the company. A score built on one factor is fragile.

**Trap:** a big grey bar with no brown bar adds nothing. Read the brown bars.

# Pair spread: when two stocks drift apart



**SHOWS**

Spread between two related stocks (BBRI vs BMRI) in  $\sigma$  away from its 60-day average.

**READ IT**

Half-life 14 days: a gap tends to halve in about three weeks. Past  $\pm 2\sigma$  is stretched.

**ACT WHEN**

Near  $\pm 2\sigma$ : buy the cheap leg, short or underweight the rich one. Now 0.51: no entry.

**Trap:** pairs can break for good (a merger, a scandal). Put a stop on the spread itself.

# Ideas

*What does the formula say to buy today?*

Stock picks for V1+ · V2 · V3 · Golden v2+ · Golden A+

# Stock picks: what the formula says today

## SHOWS

Pick a formula at the top. Each row: status, ranks, stop, risk to stop, P(win), EV, Kelly size.

## READ IT

BEAR: V3 targets 0% alpha, 20% defensive, 80% cash. WAIT rows say what each name lacks.

## ACT WHEN

Only HELD and BUY rows are actions. Size with Kelly: about 1.1% of capital at risk each.

**Trap:** “no buys today” is a valid answer. Sitting out bad markets is part of the edge.

Stock picks · V3 signal99 stocks scanned · as of 06 Oct 2026 · rules, track record and research in the Backtest tab

V1+V2V3Golden v2+Golden A+

V3: V2 + self-learning model rank ≥ 60% + no unusual jump in the last 5 days (recommended)

BEAR

Target now: 0% alpha · 20% defensive · 80% cash. Slots used: alpha 0/0, defensive 2/2. No new buys today, the rows below show what each candidate is waiting for. v3 record: 45 trades, 69% win, +2.1% per trade after costs.

| Status | Ticker | Sleeve | Score rank | Model rank | Last   | Stop (−2.5×ATR) | Break-even after | Risk to stop | Why                                                                                                                           | P(win) | EV to stop | Kelly size |
|--------|--------|--------|------------|------------|--------|-----------------|------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|--------|------------|------------|
| HELD   | INDF   | DEF    | 100%       | 65%        | 6,850  | 6,553           | 7,060            | −4.3%        | bought 22 Sep 2026 at 6,901 · −1.2% · stop 6,553                                                                              | 68%    | +2.2%      | 20%        |
| HELD   | MYOR   | DEF    | 95%        | 95%        | 1,405  | 1,382           | 1,403            | −1.6%        | bought 05 Oct 2026 at 1,370 · +2.0% · stop 1,382                                                                              | 68%    | +3.0%      | 20%        |
| WAIT   | MMIX   | ALPHA  | 100%       | 6%         | 935    | 853             | 968              | −8.8%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.8%      | 13%        |
| WAIT   | UVCR   | ALPHA  | 99%        | 4%         | 284    | 227             | 307              | −20.1%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | −2.9%      | 5%         |
| WAIT   | SGER   | ALPHA  | 98%        | 8%         | 775    | 685             | 811              | −11.6%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | −0.1%      | 9%         |
| WAIT   | SMIL   | ALPHA  | 96%        | 7%         | 472    | 428             | 489              | −9.2%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.6%      | 12%        |
| WAIT   | AYAM   | ALPHA  | 95%        | 12%        | 530    | 486             | 548              | −8.4%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.9%      | 13%        |
| WAIT   | ERAA   | ALPHA  | 94%        | 13%        | 625    | 563             | 650              | −10.0%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.4%      | 11%        |
| WAIT   | TINS   | ALPHA  | 93%        | 31%        | 4,670  | 4,183           | 4,865            | −10.4%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.2%      | 11%        |
| WAIT   | BFIN   | ALPHA  | 92%        | 44%        | 925    | 849             | 955              | −8.2%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.9%      | 13%        |
| WAIT   | SSIA   | ALPHA  | 90%        | 11%        | 1,760  | 1,583           | 1,831            | −10.0%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.4%      | 11%        |
| WAIT   | PTBA   | ALPHA  | 89%        | 24%        | 3,400  | 3,109           | 3,516            | −8.6%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.8%      | 13%        |
| WAIT   | AADI   | ALPHA  | 88%        | 14%        | 12,125 | 10,920          | 12,607           | −9.9%        | alpha sleeve is only used in a bull market                                                                                    | 68%    | +0.4%      | 11%        |
| WAIT   | PTRO   | ALPHA  | 87%        | 40%        | 5,100  | 4,493           | 5,343            | −11.9%       | alpha sleeve is only used in a bull market                                                                                    | 68%    | −0.2%      | 9%         |
| WAIT   | ITMG   | DEF    | 99%        | 38%        | 26,100 | 24,631          | 26,688           | −5.6%        | RSI 56 ≥ 40 (wait for pullback) · model rank 38% < 60%                                                                        | 68%    | +1.8%      | 20%        |
| WAIT   | MMIX   | DEF    | 98%        | 6%         | 935    | 853             | 968              | −8.8%        | foreign crowded 6% · RSI 85 ≥ 40 (wait for pullback) · model rank 6% < 60%                                                    | 68%    | +0.8%      | 13%        |
| WAIT   | UNVR   | DEF    | 96%        | 81%        | 1,595  | 1,533           | 1,620            | −3.9%        | volume spike 2.3×                                                                                                             | 68%    | +2.3%      | 20%        |
| WAIT   | CMRY   | DEF    | 94%        | 56%        | 4,420  | 4,191           | 4,511            | −5.2%        | RSI 46 ≥ 40 (wait for pullback) · model rank 56% < 60%                                                                        | 68%    | +1.9%      | 20%        |
| WAIT   | ICBP   | DEF    | 93%        | 52%        | 6,800  | 6,340           | 6,984            | −6.8%        | RSI 43 ≥ 40 (wait for pullback) · model rank 52% < 60%                                                                        | 68%    | +1.4%      | 16%        |
| WAIT   | BBCA   | DEF    | 92%        | 85%        | 6,100  | 5,850           | 6,200            | −4.1%        | RSI 41 ≥ 40 (wait for pullback)                                                                                               | 68%    | +2.3%      | 20%        |
| WAIT   | AKRA   | DEF    | 90%        | 30%        | 1,520  | 1,415           | 1,562            | −6.9%        | RSI 62 ≥ 40 (wait for pullback) · model rank 30% < 60%                                                                        | 68%    | +1.4%      | 16%        |
| WAIT   | PGAS   | DEF    | 89%        | 74%        | 1,420  | 1,334           | 1,454            | −6.0%        | RSI 41 ≥ 40 (wait for pullback)                                                                                               | 68%    | +1.6%      | 18%        |
| WAIT   | UNTR   | DEF    | 88%        | 32%        | 26,000 | 23,888          | 26,845           | −8.1%        | volume spike 2.4× · foreign crowded 3% · RSI 56 ≥ 40 (wait for pullback) · model rank 32% < 60% · unusual jump in last 5 days | 68%    | +1.0%      | 14%        |
| WAIT   | ASII   | DEF    | 87%        | 27%        | 4,820  | 4,556           | 4,926            | −5.5%        | RSI 53 ≥ 40 (wait for pullback) · model rank 27% < 60%                                                                        | 68%    | +1.8%      | 20%        |

Sizing with Kelly, Bayes and EV.

P(win) is the Bayes estimate of V3’s win rate from its 45 closed trades (69% observed, pulled slightly toward 50%). EV to stop assumes the full loss to the stop when wrong, so it is a cautious estimate. Kelly size = ¼ Kelly with a cautious win rate (22%) + 10 positions = 1.1% of capital at risk per stock, divided by that stock’s distance to its stop, capped at 20%. A wide stop gets a smaller position; a tight stop gets a larger one, so each trade risks the same amount. Stop levels are estimated from today’s close. Full method in Backtest → Kelly · Bayes · Expected value. Not investment advice.

# Allocation

*How much should go into each stock?*

Allocation plan

Efficient frontier

# Allocation plan: from ideas to lots

Allocation plan

size = conviction<sup>2</sup> ÷ volatility ÷ (1 + correlation), capped by 1.5% loss-at-stop, 15% max weight and 10% of daily value · exposure set by market regime

Capital IDR

472000000

ideas

max loss at stop / name

1.5%

exposure

auto (regime)

ADD TICKER

+ Add

Copy orders

↗ Share

Regime: CAUTIOUS (score -2) → target 60% invested, 40% cash. – breadth 28% of stocks above MA50 · – IHSG below its MA50 · + outlook 57% chance IHSG higher in 20D · index volatility 15% vs 16% (60D) · – foreign net selling IDR 12.54 tn (20D).

Core weights: AADI 14.4%, ANTM 12.0%, PNLF 7.5%. Buy/add: WIFI 119 lots, PNLF 1,787 lots, ACES 1,092 lots, MYOR 251 lots, ELSA 533 lots. Reduce/exit: AADI 44 lots, ANTM 126 lots, BBRI exit (stop broken), GOTO exit (stop broken), TLKM exit (stop broken). If every stop were hit the book would lose **IDR 16.4 mn** (3.5% of capital); a 1-in-20 bad day costs about IDR 8.0 mn. Model output for a disciplined process, not investment advice; orders are rounded down to whole lots.

Invested

59.8%

IDR 282.1 mn

Cash

40.2%

IDR 189.9 mn

Names

7

effective 6.1

Portfolio vol

15.7%

annualised, 239D

Beta

0.45

vs IHSG

Loss if all stops hit

3.5%

IDR 16.4 mn

VaR 1D 95%

1.70%

IDR 8.0 mn

Diversification

1.74×

sum of vols ÷ book vol

| Ticker | Source  | Category                                                      | Conviction | Trend       | Last | Stop   | Target | R:R                    | P(stop first) | P(target first) | Vol | Now  | Target | Lots  | Action | Risk at stop |              |
|--------|---------|---------------------------------------------------------------|------------|-------------|------|--------|--------|------------------------|---------------|-----------------|-----|------|--------|-------|--------|--------------|--------------|
| AADI   | held    | Defensive Compounders<br>alpha (t 1.4) · low beta             | 1.13       | <div></div> | 4/4  | 12,125 | 11,650 | 12,150                 | 0.1           | 27%             | 73% | 43%  | 25.7%  | 14.4% | 56     | TRIM -44     | IDR 2.7 mn × |
| ANTM   | held    | Aggressive Leaders<br>beta-driven · market beta               | 1.00       | <div></div> | 4/4  | 3,250  | 3,070  | 3,610<br>2R from trail | 2.0           | 44%             | 25% | 38%  | 20.7%  | 12.0% | 174    | TRIM -126    | IDR 3.1 mn × |
| PNLF   | idea #2 | Defensive Compounders<br>beta-driven · market beta            | 0.84       | <div></div> | 0/4  | 198    | 186    | 222                    | 2.0           | 33%             | 13% | 29%  | 0.0%   | 7.5%  | 1,787  | BUY +1,787   | IDR 2.1 mn × |
| ACES   | idea #3 | Defensive Laggards<br>beta-driven · low beta                  | 0.69       | <div></div> | 0/4  | 324    | 306    | 360                    | 2.0           | 29%             | 6%  | 23%  | 0.0%   | 7.5%  | 1,092  | BUY +1,092   | IDR 2.0 mn × |
| MYOR   | idea #4 | Defensive Laggards<br>beta-driven · low beta                  | 0.68       | <div></div> | 0/4  | 1,405  | 1,330  | 1,555                  | 2.0           | 40%             | 13% | 25%  | 0.0%   | 7.5%  | 251    | BUY +251     | IDR 1.9 mn × |
| ELSA   | idea #5 | Defensive Compounders<br>beta-driven · low beta               | 0.81       | <div></div> | 0/4  | 645    | 595    | 745                    | 2.0           | 39%             | 15% | 42%  | 0.0%   | 7.3%  | 533    | BUY +533     | IDR 2.7 mn × |
| WIFI   | idea #1 | High-Beta Laggards<br>beta-driven · high beta                 | 0.71       | <div></div> | 1/4  | 1,450  | 1,285  | 1,780                  | 2.0           | 40%             | 15% | 64%  | 0.0%   | 3.7%  | 119    | BUY +119     | IDR 2.0 mn × |
| BBRI   | held    | Defensive Compounders<br>beta-driven · market beta            | 0.82       | <div></div> | 1/4  | 3,110  | 3,220  | 3,360                  | –             | –               | –   | 24%  | 32.9%  | 0.0%  | 0      | EXIT         | – ×          |
| GOTO   | held    | Defensive Laggards<br>negative alpha · low beta · ▲ vol spike | 0.20       | <div></div> | 0/4  | 31     | 59     | 70                     | –             | –               | –   | 111% | 1.3%   | 0.0%  | 0      | EXIT         | – ×          |
| TLKM   | held    | Defensive Laggards<br>beta-driven · market beta               | 0.67       | <div></div> | 0/4  | 2,280  | 2,910  | 3,240                  | –             | –               | –   | 30%  | 19.3%  | 0.0%  | 0      | EXIT         | – ×          |
| CASH   |         | buffer set by regime + lot roundina                           |            |             |      |        |        |                        |               |                 |     |      |        | 40.2% |        |              | IDR 189.9 mn |

## SHOWS

Size = conviction<sup>2</sup> × trend ÷ volatility ÷ (1 + correlation), capped by 1.5% loss at stop.

## READ IT

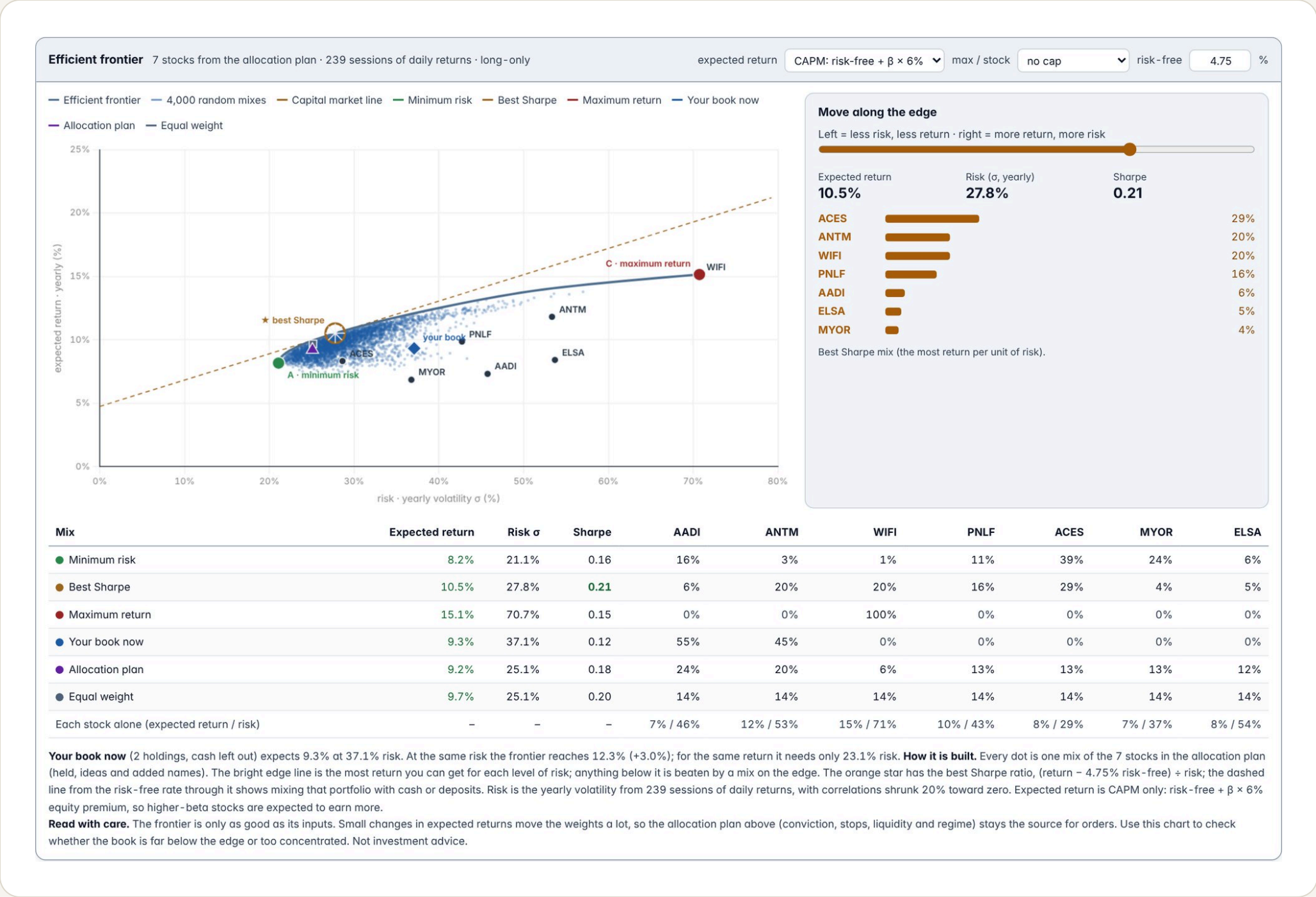
CAUTIOUS regime → 60% invested. Each row ends in an action: TRIM, BUY +lots or EXIT.

## ACT WHEN

Do EXITs and TRIMs first, then buys. If every stop hits, the loss is IDR 16.4M (3.5%).

**Trap:** orders round to whole lots, so small accounts can’t hit every target weight.

# Efficient frontier: the best mix for each risk



## SHOWS

4,000 random mixes of the plan’s 7 stocks (dots). The curve: best return for each risk.

## READ IT

Anything below the curve can be beaten. Your book: 9.3% expected return at 37.1% risk.

## ACT WHEN

Move toward the curve: at the same risk the frontier offers 12.3%, or 9.3% at 23% risk.

**Trap:** it is only as good as the expected returns fed in. Small changes swing the weights.

# Backtest

*Would these rules have worked, and is the edge real?*

Formula comparison

Kelly · Bayes · EV

Institutional signal v3

Pattern lab

Backtest by segment

Model backtest

# Formula comparison: same stocks, same window

SHOWS

Each formula’s growth, 17 Apr–6 Oct 2026, plus trades, win rate, profit factor, drawdown.

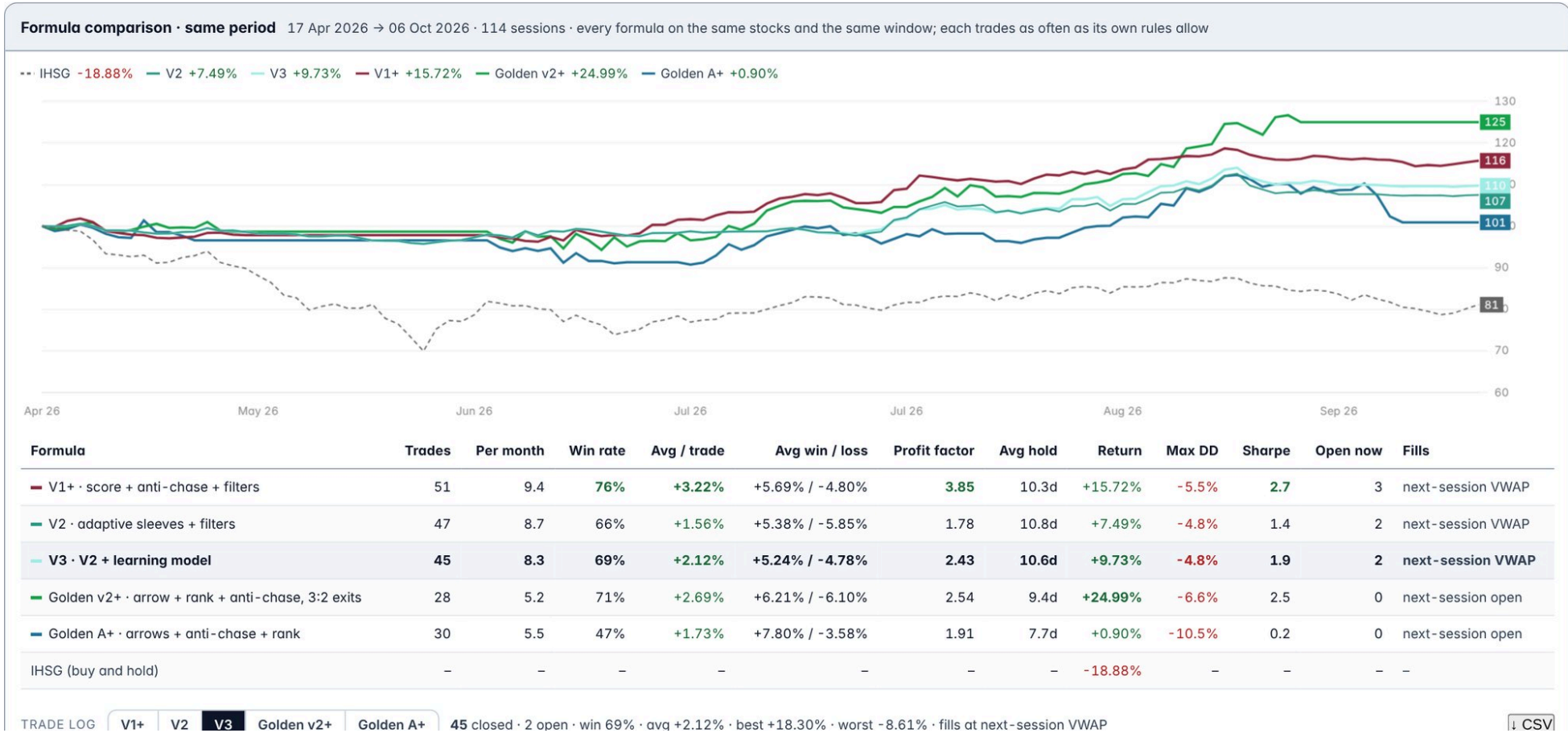
READ IT

All beat IHSG (–18.9%). V1+: 51 trades, 76% win, +3.22% a trade, profit factor 3.85.

ACT WHEN

Compare average per trade and drawdown, not total return: trade counts differ.

**Trap:** the rules were chosen after seeing this year. Expect weaker live results.



# Kelly, Bayes, EV: is the edge real, how big a bet?

SHOWS

EV per trade, Kelly bet size and a Bayes range for the true win rate, for every formula.

READ IT

V3: EV +2.12% a trade. Kelly 41% → cautious 22%, over 10 slots ≈ 1.1% at risk each.

ACT WHEN

Trust a formula when EV > 0 and the low end of its Bayes range stays above 50%.

Trap: full Kelly assumes you know the true odds. You don’t, so use a fraction.

Kelly • Bayes • Expected value17 Apr 2026 → 06 Oct 2026 • trade returns after costs • cards show the formula selected in Ideas (V3)

Expected value

EV = (P<sub>win</sub> × W) – (P<sub>loss</sub> × L)

= (0.69 × 5.24%) – (0.31 × 4.78%)

= **+2.12%** per trade

Average profit you should expect from each V3 trade. Above 0 means the formula makes money over many trades even if single trades lose. At 8.3 trades a month and 10% per position, that is about **+1.76%** of capital a month.

Kelly criterion

f = p – q / b

b = W / L = 5.24% / 4.78% = 1.10

f = 0.69 – 0.31 / 1.10 = **41%** of capital at risk

cautious p (10th pct) 0.59 → f = **22%**

Kelly is the share of capital to put at risk (entry to stop) for the fastest long-run growth. It assumes the win rate is known exactly, so we use the cautious p, take ½ Kelly and split it across 10 positions: **1.1% risk per position**.

Bayes theorem

P(A|B) = P(B|A) × P(A) / P(B)

A = beat the market over 10 sessions, B = “Unusual drop (return < –3σ)”

= 1.3% × 46.4% / 0.94% = **63.7%**

prior 46.4% → posterior 63.7% • likelihood ratio 2.03

Start from the base rate, then update it with the evidence. A likelihood ratio above 1 raises the chance and below 1 lowers it. Below, the win rate of each formula is also a Bayes posterior: it starts from no opinion and is updated with each closed trade.

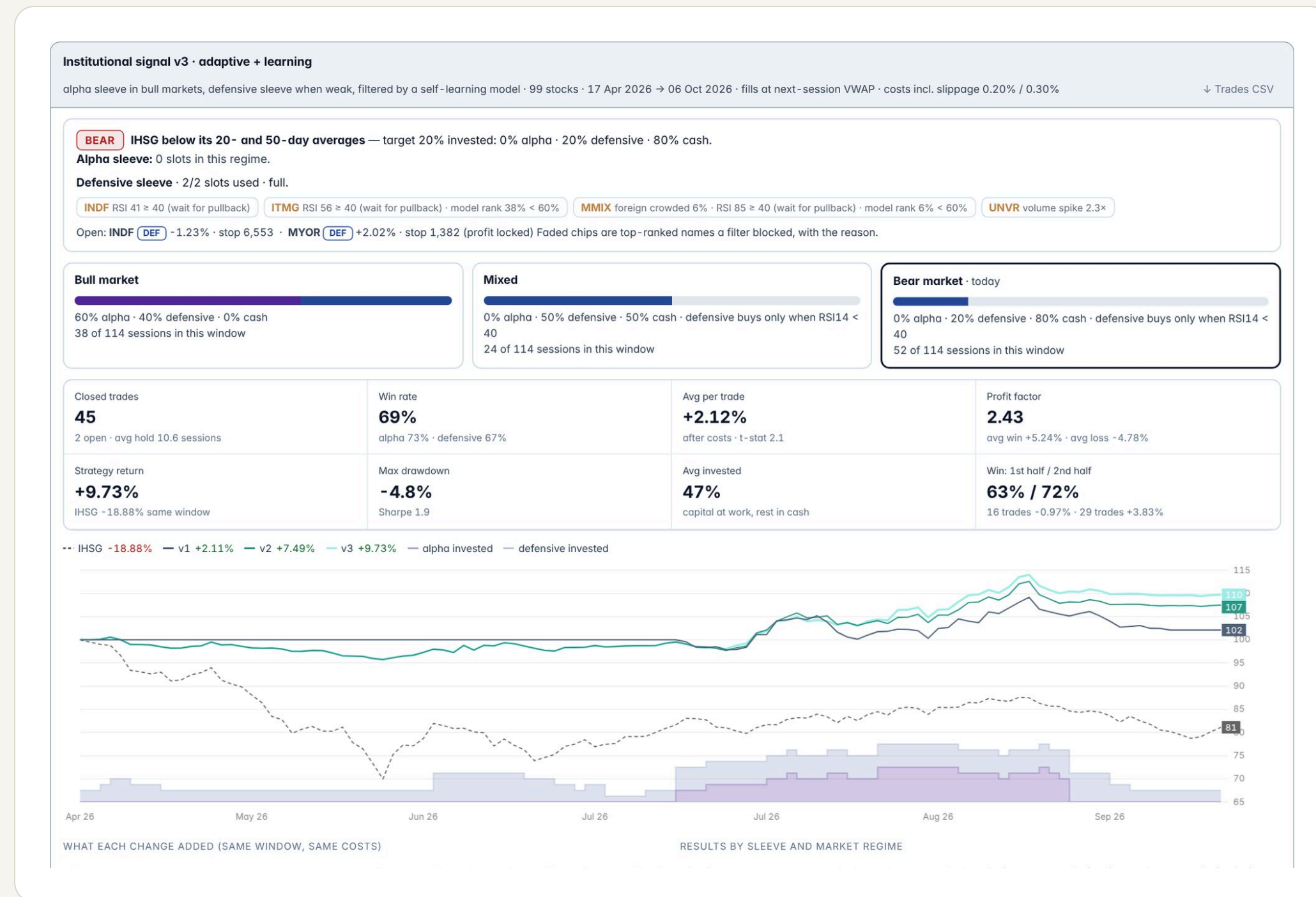
Expected value and Kelly sizing • every formula, same window

| Formula                                           | Closed trades | Win p | Bayes p (90% band) | Avg win W | Avg loss L | Payoff b | EV / trade | EV / month* | Kelly f | Kelly, cautious p | Risk / position | Verdict       |
|---------------------------------------------------|---------------|-------|--------------------|-----------|------------|----------|------------|-------------|---------|-------------------|-----------------|---------------|
| V2 • adaptive sleeves + filters                   | 47            | 66%   | 65% (54%–76%)      | +5.38%    | -5.85%     | 0.92     | +1.56%     | +1.35%      | 29%     | 9%                | 0.5%            | POSITIVE EDGE |
| V3 • V2 + learning model                          | 45            | 69%   | 68% (57%–79%)      | +5.24%    | -4.78%     | 1.10     | +2.12%     | +1.76%      | 41%     | 22%               | 1.1%            | POSITIVE EDGE |
| V1+ • score + anti-chase + filters                | 51            | 76%   | 75% (65%–85%)      | +5.69%    | -4.80%     | 1.18     | +3.22%     | +3.02%      | 57%     | 40%               | 2.0%            | POSITIVE EDGE |
| Golden v2+ • arrow + rank + anti-chase, 3:2 exits | 28            | 71%   | 70% (56%–83%)      | +6.21%    | -6.10%     | 1.02     | +2.69%     | +1.39%      | 43%     | 19%               | 0.9%            | POSITIVE EDGE |
| Golden A+ • arrows + anti-chase + rank            | 30            | 47%   | 47% (33%–61%)      | +7.80%    | -3.58%     | 2.18     | +1.73%     | +0.96%      | 22%     | 6%                | 0.3%            | POSITIVE EDGE |

Bayes • how much each pattern changes the chance of beating the market over 10 sessions

| Pattern (B)                         | Events | P(B   win) | P(B   loss) | Likelihood ratio | Prior P(win) | Posterior P(win   B) | Change | Shrunk posterior (90% band) | Read   |
|-------------------------------------|--------|------------|-------------|------------------|--------------|----------------------|--------|-----------------------------|--------|
| Unusual drop (return < –3σ)         | 135    | 1.29%      | 0.64%       | 2.03             | 46.4%        | 63.7%                | +17.3% | 53.4% (49%–58%)             | RAISES |
| Big orders on a down day            | 193    | 1.64%      | 1.09%       | 1.50             | 46.4%        | 56.5%                | +10.1% | 51.4% (47%–55%)             | RAISES |
| Big orders: ticket size ≥ 2× normal | 461    | 3.89%      | 2.63%       | 1.48             | 46.4%        | 56.2%                | +9.8%  | 53.2% (50%–56%)             | RAISES |

# Institutional signal v3: the adaptive model



## SHOWS

Today's regime and sleeves, headline stats, and growth of each version vs IHSG.

## READ IT

BEAR today: 0% alpha, 20% defensive, 80% cash. 45 trades, 69% win, +9.73% vs -18.88%.

## ACT WHEN

Follow the regime's cash target: in BEAR the model keeps most money out of stocks.

**Trap:** much of v3's edge is simply being in cash during bad weeks. Check the shading.

# Pattern lab: which patterns paid

| Pattern lab & learning model 99 stocks · patterns from 12 Jan 2026 · model retrained every session on the previous 120 sessions (114 refits) |                                     |            |                                                                         |          |            |              |        |             |          |                  |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------|-------------------------------------------------------------------------|----------|------------|--------------|--------|-------------|----------|------------------|
| SCANNER TODAY · RANKED BY THE LEARNING MODEL (CLICK A ROW TO OPEN THE STOCK)                                                                 |                                     |            |                                                                         |          |            |              |        |             |          |                  |
| #                                                                                                                                            | Ticker                              | Model rank | Patterns active now (green = historically good, red = historically bad) |          |            | Ticket trend |        | Foreign 60D |          | vs 20D VWAP      |
| 1                                                                                                                                            | SMRA                                | 100%       | -                                                                       |          |            | +15.78%      |        | -13.47%     |          | -14.49%          |
| 2                                                                                                                                            | BRIS                                | 99%        | <div>Unusual drop</div> <div>Volume spike 2.2×</div>                    |          |            | -9.42%       |        | -6.57%      |          | -9.34%           |
| 3                                                                                                                                            | BSDE                                | 98%        | <div>Big orders</div> <div>Volume spike 2.1×</div>                      |          |            | +56.62%      |        | -16.69%     |          | -5.96%           |
| 4                                                                                                                                            | KPIG                                | 96%        | -                                                                       |          |            | +12.70%      |        | -8.69%      |          | -34.34%          |
| 5                                                                                                                                            | MYOR                                | 95%        | -                                                                       |          |            | +5.23%       |        | -4.07%      |          | -4.74%           |
| 6                                                                                                                                            | JPFA                                | 94%        | -                                                                       |          |            | +1.99%       |        | -18.52%     |          | -8.32%           |
| 7                                                                                                                                            | ELSA                                | 93%        | -                                                                       |          |            | +17.23%      |        | -2.70%      |          | -6.41%           |
| 8                                                                                                                                            | TLKM                                | 92%        | <div>Volume spike 2.6×</div>                                            |          |            | -6.45%       |        | -9.47%      |          | -6.42%           |
| 9                                                                                                                                            | CPIN                                | 90%        | -                                                                       |          |            | +15.79%      |        | -25.17%     |          | -5.59%           |
| 10                                                                                                                                           | LSIP                                | 89%        | <div>Foreign crowded 20D</div>                                          |          |            | +6.16%       |        | +2.50%      |          | -8.01%           |
| 11                                                                                                                                           | PANI                                | 88%        | <div>Unusual jump</div>                                                 |          |            | -5.74%       |        | -3.64%      |          | -7.32%           |
| 12                                                                                                                                           | BBNI                                | 87%        | -                                                                       |          |            | -5.92%       |        | -2.59%      |          | -4.08%           |
| 13                                                                                                                                           | INET                                | 86%        | -                                                                       |          |            | +33.69%      |        | -0.00%      |          | -12.60%          |
| PATTERN SCORECARD · WHAT HAPPENED OVER THE NEXT 10 SESSIONS (EXCESS RETURN VS THE AVERAGE LIQUID STOCK)                                      |                                     |            |                                                                         |          |            |              |        |             |          |                  |
| Type                                                                                                                                         | Pattern                             | Events     | 10D excess                                                              | Hit rate | t (by day) | Bull         | Bear   | 1st half    | 2nd half | Verdict          |
| Big money                                                                                                                                    | Big orders: ticket size ≥ 2× normal | 461        | +1.17%                                                                  | 56%      | 1.6        | +1.07%       | +1.25% | +1.24%      | +1.13%   | EDGE             |
| Big money                                                                                                                                    | Big orders on a down day            | 193        | +1.56%                                                                  | 56%      | 1.4        | +0.78%       | +1.97% | +2.47%      | +0.90%   | EDGE             |
| Unusual                                                                                                                                      | Unusual drop (return < -3σ)         | 135        | +2.30%                                                                  | 64%      | 0.8        | -0.00%       | +2.45% | +2.95%      | +0.13%   | EDGE             |
| Unusual                                                                                                                                      | Unusual jump (return > +3σ)         | 140        | -1.76%                                                                  | 36%      | -1.2       | +0.89%       | -2.71% | -0.71%      | -3.76%   | AVOID            |
| Unusual                                                                                                                                      | Range expansion (> 2.5×ATR)         | 313        | +0.43%                                                                  | 50%      | -0.2       | +0.46%       | +0.41% | +0.68%      | -0.10%   | NO RELIABLE EDGE |
| Volume                                                                                                                                       | Volume spike ≥ 2× on up day         | 736        | -0.38%                                                                  | 44%      | -0.7       | +1.24%       | -1.98% | -0.20%      | -0.58%   | BULL ONLY        |
| Volume                                                                                                                                       | Volume spike ≥ 2× on down day       | 465        | +0.21%                                                                  | 49%      | -1.5       | -0.75%       | +0.68% | +0.82%      | -0.98%   | NO RELIABLE EDGE |

## SHOWS

Top: today’s stocks ranked by the learning model. Bottom: each pattern’s 10-day excess return.

## READ IT

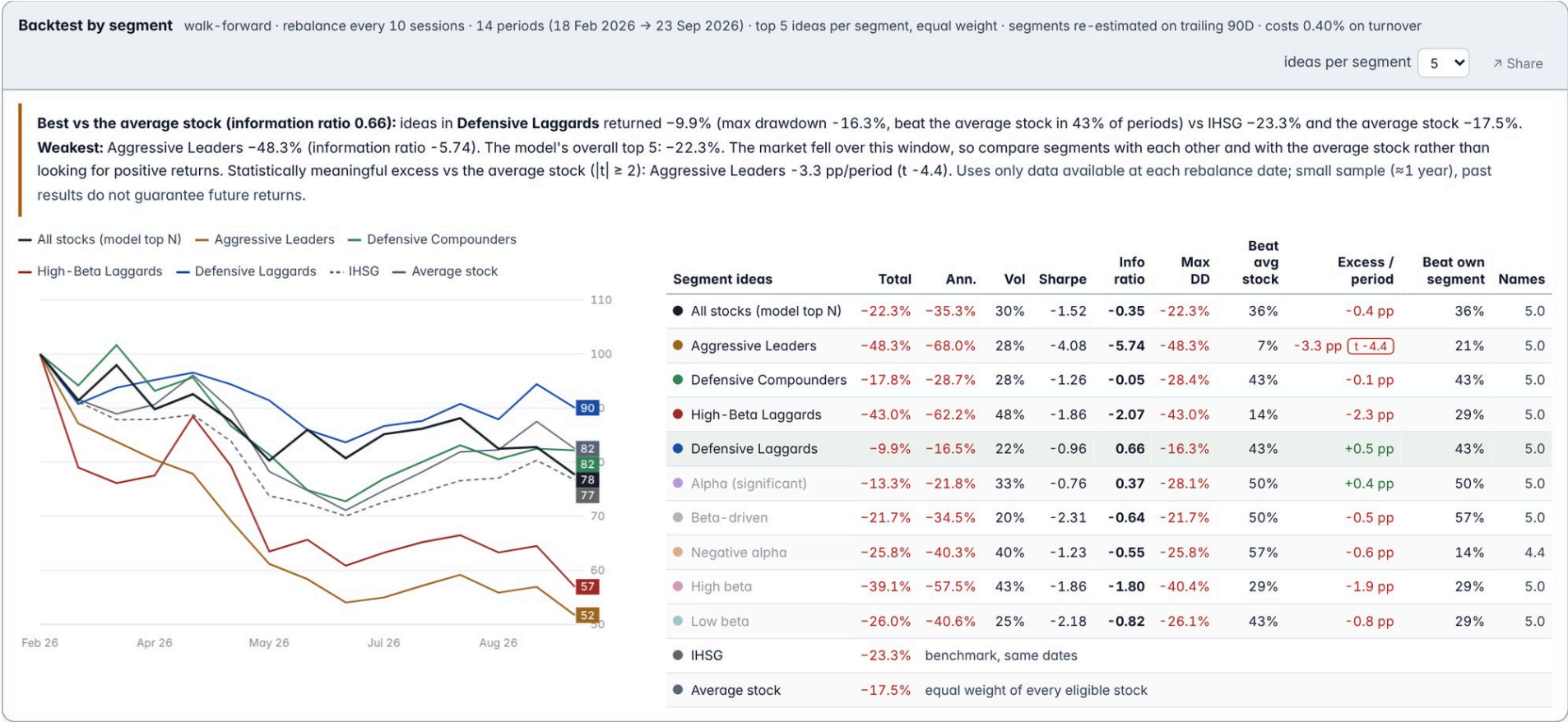
EDGE = paid in both halves of the year. AVOID = reliably lost, e.g. unusual jump (−1.76%).

## ACT WHEN

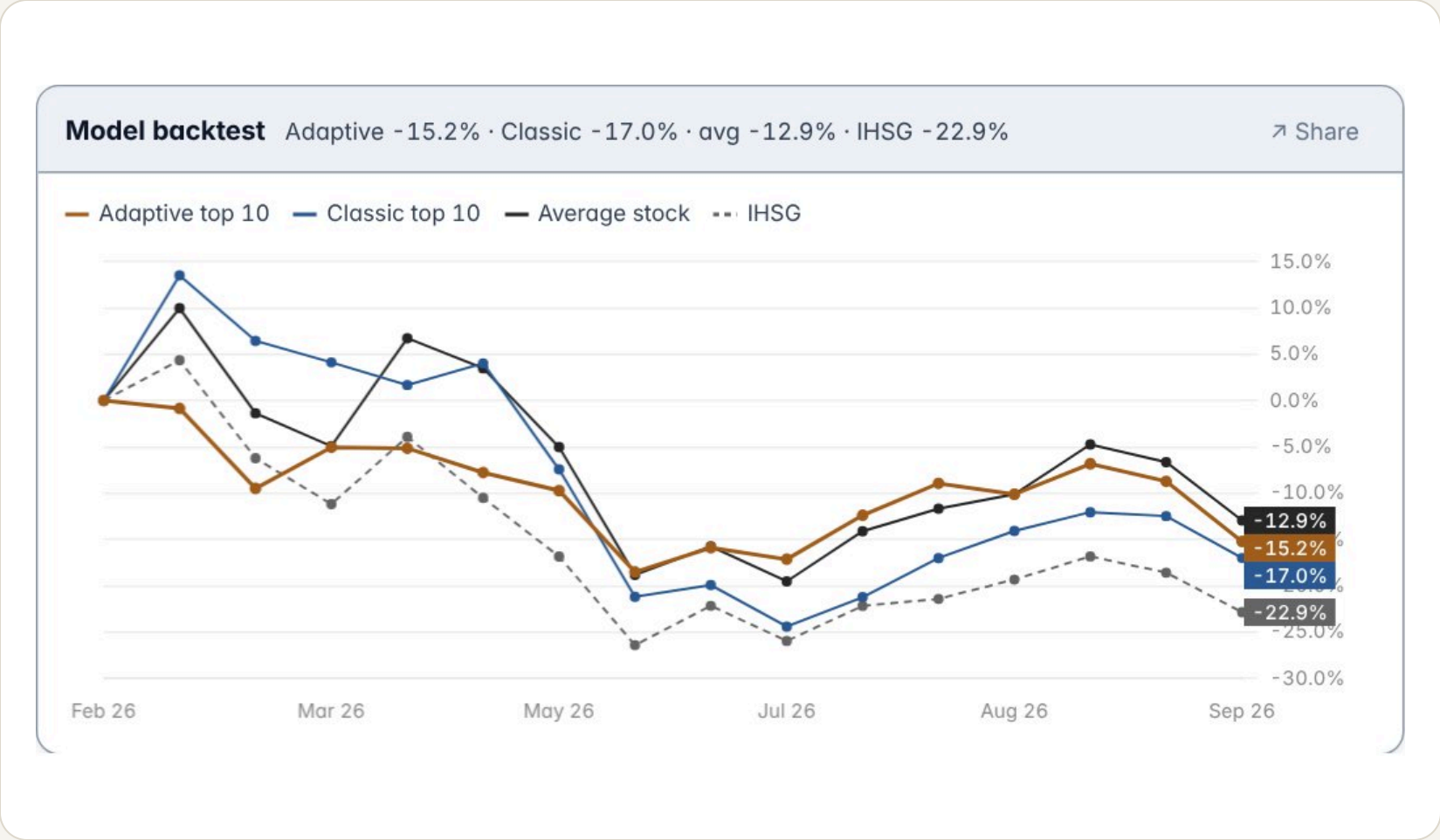
Favour names with green patterns active; treat red ones (unusual jump) as a warning.

Trap: few events and a small t-stat can look great by luck. Check the count first.

# Backtest by segment: which kind of stock worked



# Model backtest and what the data rewards



**Model backtest:** top 10 by the adaptive and classic models vs the average stock and IHSG. Both beat IHSG but lagged the average stock.



**What the data rewards:** rank IC per factor over the next 10 days. Green predicted gains; red predicted the opposite.

**Trap:** here low volatility paid and momentum failed. What the market rewards changes over time.

PUTTING IT TOGETHER

# A five-minute daily reading routine

1

## Market

Regime, breadth and foreign flow. Is the tide in or out?

2

## Portfolio

vs IHSG, any stop hit, the weakest rows.

3

## Risk

VaR breaches, and risk share above weight.

4

## Act

Ideas, then  
Allocation: exits first, buys sized by the plan.

5

## Audit

Weekly: is the backtest edge still holding?

Outside in, then inside out: environment → what I own → what could hurt → what to do → does the method still work.

CLASS EXERCISE · 20 MINUTES

# Read the charts, then make a call

## 1 · P&L vs VaR

With 239 days and a 95% VaR, how many breaches should we expect? Is 12 a problem?

## 2 · Risk contribution

Which holding would you trim first to calm the portfolio, and why?

## 3 · Breadth + foreign flow

Is this a broad or a narrow market? Who is selling, and which stocks feel it most?

## 4 · Efficient frontier

How much expected return is the current book leaving on the table at its risk level?

Answers are in the speaker notes · Use the four questions: axes, benchmark, now, decision